

FANUC Robot series

R-30iA/R-30iB/R-30iB Plus CONTROLLER

INTEGRATED WELD CONTROL INTERFACE
(DENGSHA SPECIFICATIONS)

OPERATOR' S MANUAL

MAROCDENG04071E REV. D

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B-82624EN/03.01

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FANUC conducts courses on its systems and products on a regularly scheduled basis at the company's world headquarters in Rochester Hills, Michigan. For additional information contact

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For customer assistance, including Technical Support, Service, Parts & Part Repair, and Marketing Requests, contact the Customer Resource Center, 24 hours a day, at 1-800-47-ROBOT (1-800-477-6268). International customers should call 011-1-248-377-7159.

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Patents

One or more of the following U.S. patents might be related to the FANUC products described in this manual.

FANUC America Corporation Patent List

4,630,567 4,639,878 4,707,647 4,708,175 4,708,580 4,942,539 4,984,745
5,238,029 5,239,739 5,272,805 5,293,107 5,293,911 5,331,264 5,367,944
5,373,221 5,421,218 5,434,489 5,644,898 5,670,202 5,696,687 5,737,218
5,823,389 5,853,027 5,887,800 5,941,679 5,959,425 5,987,726 6,059,092
6,064,168 6,070,109 6,086,294 6,122,062 6,147,323 6,204,620 6,243,621
6,253,799 6,285,920 6,313,595 6,325,302 6,345,818 6,356,807 6,360,143
6,378,190 6,385,508 6,425,177 6,477,913 6,490,369 6,518,980 6,540,104
6,541,757 6,560,513 6,569,258 6,612,449 6,703,079 6,705,361 6,726,773
6,768,078 6,845,295 6,945,483 7,149,606 7,149,606 7,211,978 7,266,422
7,399,363

FANUC CORPORATION Patent List

4,571,694 4,626,756 4,700,118 4,706,001 4,728,872 4,732,526 4,742,207
4,835,362 4,894,596 4,899,095 4,920,248 4,931,617 4,934,504 4,956,594
4,967,125 4,969,109 4,970,370 4,970,448 4,979,127 5,004,968 5,006,035
5,008,834 5,063,281 5,066,847 5,066,902 5,093,552 5,107,716 5,111,019
5,130,515 5,136,223 5,151,608 5,170,109 5,189,351 5,267,483 5,274,360
5,292,066 5,300,868 5,304,906 5,313,563 5,319,443 5,325,467 5,327,057
5,329,469 5,333,242 5,337,148 5,371,452 5,375,480 5,418,441 5,432,316
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5,691,615 5,698,121 5,708,342 5,715,375 5,719,479 5,727,132 5,742,138
5,742,144 5,748,854 5,749,058 5,760,560 5,773,950 5,783,922 5,799,135
5,812,408 5,841,257 5,845,053 5,872,894 5,887,122 5,911,892 5,912,540
5,920,678 5,937,143 5,980,082 5,983,744 5,987,591 5,988,850 6,023,044
6,032,086 6,040,554 6,059,169 6,088,628 6,097,169 6,114,824 6,124,693
6,140,788 6,141,863 6,157,155 6,160,324 6,163,124 6,177,650 6,180,898
6,181,096 6,188,194 6,208,105 6,212,444 6,219,583 6,226,181 6,236,011
6,236,896 6,250,174 6,278,902 6,279,413 6,285,921 6,298,283 6,321,139
6,324,443 6,328,523 6,330,493 6,340,875 6,356,671 6,377,869 6,382,012
6,384,371 6,396,030 6,414,711 6,424,883 6,431,018 6,434,448 6,445,979
6,459,958 6,463,358 6,484,067 6,486,629 6,507,165 6,654,666 6,665,588
6,680,461 6,696,810 6,728,417 6,763,284 6,772,493 6,845,296 6,853,881
6,888,089 6,898,486 6,917,837 6,928,337 6,965,091 6,970,802 7,038,165
7,069,808 7,084,900 7,092,791 7,133,747 7,143,100 7,149,602 7,131,848
7,161,321 7,171,041 7,174,234 7,173,213 7,177,722 7,177,439 7,181,294
7,181,313 7,280,687 7,283,661 7,291,806 7,299,713 7,315,650 7,324,873
7,328,083 7,330,777 7,333,879 7,355,725 7,359,817 7,373,220 7,376,488
7,386,367 7,464,623 7,447,615 7,445,260 7,474,939 7,486,816 7,495,192
7,501,778 7,502,504 7,508,155 7,512,459 7,525,273 7,526,121

Conventions

WARNING

Information appearing under the "WARNING" caption concerns the protection of personnel. It is boxed and bolded to set it apart from the surrounding text.

CAUTION

Information appearing under the "CAUTION" caption concerns the protection of equipment, software, and data. It is boxed and bolded to set it apart from the surrounding text.

Note Information appearing next to NOTE concerns related information or useful hints.

- **Original Instructions**

Before using the Robot, be sure to read the "FANUC Robot Safety Manual (B-80687EN)" and understand the content.

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Should you wish to export or re-export these products, please contact FANUC for advice.

In this manual we have tried as much as possible to describe all the various matters.

However, we cannot describe all the matters which must not be done, or which cannot be done, because there are so many possibilities.

Therefore, matters which are not especially described as possible in this manual should be regarded as "impossible".

Safety

FANUC America Corporation is not and does not represent itself as an expert in safety systems, safety equipment, or the specific safety aspects of your company and/or its work force. It is the responsibility of the owner, employer, or user to take all necessary steps to guarantee the safety of all personnel in the workplace.

The appropriate level of safety for your application and installation can be best determined by safety system professionals. FANUC America Corporation therefore, recommends that each customer consult with such professionals in order to provide a workplace that allows for the safe application, use, and operation of FANUC America Corporation systems.

According to the industry standard ANSI/RIA R15-06, the owner or user is advised to consult the standards to ensure compliance with its requests for Robotics System design, usability, operation, maintenance, and service. Additionally, as the owner, employer, or user of a robotic system, it is your responsibility to arrange for the training of the operator of a robot system to recognize and respond to known hazards associated with your robotic system and to be aware of the recommended operating procedures for your particular application and robot installation.

Ensure that the robot being used is appropriate for the application. Robots used in classified (hazardous) locations must be certified for this use.

FANUC America Corporation therefore, recommends that all personnel who intend to operate, program, repair, or otherwise use the robotics system be trained in an approved FANUC America Corporation training course and become familiar with the proper operation of the system. Persons responsible for programming the system—including the design, implementation, and debugging of application programs—must be familiar with the recommended programming procedures for your application and robot installation.

The following guidelines are provided to emphasize the importance of safety in the workplace.

CONSIDERING SAFETY FOR YOUR ROBOT INSTALLATION

Safety is essential whenever robots are used. Keep in mind the following factors with regard to safety:

- The safety of people and equipment
- Use of safety enhancing devices
- Techniques for safe teaching and manual operation of the robot(s)
- Techniques for safe automatic operation of the robot(s)
- Regular scheduled inspection of the robot and workcell
- Proper maintenance of the robot

Keeping People Safe

The safety of people is always of primary importance in any situation. When applying safety measures to your robotic system, consider the following:

- External devices
- Robot(s)
- Tooling
- Workpiece

Using Safety Enhancing Devices

Always give appropriate attention to the work area that surrounds the robot. The safety of the work area can be enhanced by the installation of some or all of the following devices:

- Safety fences, barriers, or chains
- Light curtains
- Interlocks
- Pressure mats
- Floor markings
- Warning lights
- Mechanical stops
- EMERGENCY STOP buttons
- DEADMAN switches

Setting Up a Safe Workcell

A safe workcell is essential to protect people and equipment. Observe the following guidelines to ensure that the workcell is set up safely. These suggestions are intended to supplement and not replace existing federal, state, and local laws, regulations, and guidelines that pertain to safety.

- Sponsor your personnel for training in approved FANUC America Corporation training course(s) related to your application. Never permit untrained personnel to operate the robots.
- Install a lockout device that uses an access code to prevent unauthorized persons from operating the robot.
- Use anti-tie-down logic to prevent the operator from bypassing safety measures.
- Arrange the workcell so the operator faces the workcell and can see what is going on inside the cell.
- Clearly identify the work envelope of each robot in the system with floor markings, signs, and special barriers. The work envelope is the area defined by the maximum motion range of the robot, including any tooling attached to the wrist flange that extend this range.

- Position all controllers outside the robot work envelope.
- Never rely on software or firmware based controllers as the primary safety element unless they comply with applicable current robot safety standards.
- Mount an adequate number of EMERGENCY STOP buttons or switches within easy reach of the operator and at critical points inside and around the outside of the workcell.
- Install flashing lights and/or audible warning devices that activate whenever the robot is operating, that is, whenever power is applied to the servo drive system. Audible warning devices shall exceed the ambient noise level at the end-use application.
- Wherever possible, install safety fences to protect against unauthorized entry by personnel into the work envelope.
- Install special guarding that prevents the operator from reaching into restricted areas of the work envelope.
- Use interlocks.
- Use presence or proximity sensing devices such as light curtains, mats, and capacitance and vision systems to enhance safety.
- Periodically check the safety joints or safety clutches that can be optionally installed between the robot wrist flange and tooling. If the tooling strikes an object, these devices dislodge, remove power from the system, and help to minimize damage to the tooling and robot.
- Make sure all external devices are properly filtered, grounded, shielded, and suppressed to prevent hazardous motion due to the effects of electro-magnetic interference (EMI), radio frequency interference (RFI), and electro-static discharge (ESD).
- Make provisions for power lockout/tagout at the controller.
- Eliminate *pinch points*. Pinch points are areas where personnel could get trapped between a moving robot and other equipment.
- Provide enough room inside the workcell to permit personnel to teach the robot and perform maintenance safely.
- Program the robot to load and unload material safely.
- If high voltage electrostatics are present, be sure to provide appropriate interlocks, warning, and beacons.
- If materials are being applied at dangerously high pressure, provide electrical interlocks for lockout of material flow and pressure.

Staying Safe While Teaching or Manually Operating the Robot

Advise all personnel who must teach the robot or otherwise manually operate the robot to observe the following rules:

- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Know whether or not you are using an intrinsically safe teach pendant if you are working in a hazardous environment.

- Before teaching, visually inspect the robot and work envelope to make sure that no potentially hazardous conditions exist. The work envelope is the area defined by the maximum motion range of the robot. These include tooling attached to the wrist flange that extends this range.
- The area near the robot must be clean and free of oil, water, or debris. Immediately report unsafe working conditions to the supervisor or safety department.
- FANUC America Corporation recommends that no one enter the work envelope of a robot that is on, except for robot teaching operations. However, if you must enter the work envelope, be sure all safeguards are in place, check the teach pendant DEADMAN switch for proper operation, and place the robot in teach mode. Take the teach pendant with you, turn it on, and be prepared to release the DEADMAN switch. Only the person with the teach pendant should be in the work envelope.

 WARNING

Never bypass, strap, or otherwise deactivate a safety device, such as a limit switch, for any operational convenience. Deactivating a safety device is known to have resulted in serious injury and death.

- Know the path that can be used to escape from a moving robot; make sure the escape path is never blocked.
- Isolate the robot from all remote control signals that can cause motion while data is being taught.
- Test any program being run for the first time in the following manner:

 WARNING

Stay outside the robot work envelope whenever a program is being run. Failure to do so can result in injury.

- Using a low motion speed, single step the program for at least one full cycle.
 - Using a low motion speed, test run the program continuously for at least one full cycle.
 - Using the programmed speed, test run the program continuously for at least one full cycle.
- Make sure all personnel are outside the work envelope before running production.

Staying Safe During Automatic Operation

Advise all personnel who operate the robot during production to observe the following rules:

- Make sure all safety provisions are present and active.

- Know the entire workcell area. The workcell includes the robot and its work envelope, plus the area occupied by all external devices and other equipment with which the robot interacts.
- Understand the complete task the robot is programmed to perform before initiating automatic operation.
- Make sure all personnel are outside the work envelope before operating the robot.
- Never enter or allow others to enter the work envelope during automatic operation of the robot.
- Know the location and status of all switches, sensors, and control signals that could cause the robot to move.
- Know where the EMERGENCY STOP buttons are located on both the robot control and external control devices. Be prepared to press these buttons in an emergency.
- Never assume that a program is complete if the robot is not moving. The robot could be waiting for an input signal that will permit it to continue its activity.
- If the robot is running in a pattern, do not assume it will continue to run in the same pattern.
- Never try to stop the robot, or break its motion, with your body. The only way to stop robot motion immediately is to press an EMERGENCY STOP button located on the controller panel, teach pendant, or emergency stop stations around the workcell.

Staying Safe During Inspection

When inspecting the robot, be sure to

- Turn off power at the controller.
- Lock out and tag out the power source at the controller according to the policies of your plant.
- Turn off the compressed air source and relieve the air pressure.
- If robot motion is not needed for inspecting the electrical circuits, press the EMERGENCY STOP button on the operator panel.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- If power is needed to check the robot motion or electrical circuits, be prepared to press the EMERGENCY STOP button, in an emergency.
- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.

Staying Safe During Maintenance

When performing maintenance on your robot system, observe the following rules:

- Never enter the work envelope while the robot or a program is in operation.
- Before entering the work envelope, visually inspect the workcell to make sure no potentially hazardous conditions exist.

- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Consider all or any overlapping work envelopes of adjoining robots when standing in a work envelope.
- Test the teach pendant for proper operation before entering the work envelope.
- If it is necessary for you to enter the robot work envelope while power is turned on, you must be sure that you are in control of the robot. Be sure to take the teach pendant with you, press the DEADMAN switch, and turn the teach pendant on. Be prepared to release the DEADMAN switch to turn off servo power to the robot immediately.
- Whenever possible, perform maintenance with the power turned off. Before you open the controller front panel or enter the work envelope, turn off and lock out the 3-phase power source at the controller.
- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.

 WARNING

Lethal voltage is present in the controller WHENEVER IT IS CONNECTED to a power source. Be extremely careful to avoid electrical shock. HIGH VOLTAGE IS PRESENT at the input side whenever the controller is connected to a power source. Turning the disconnect or circuit breaker to the OFF position removes power from the output side of the device only.

- Release or block all stored energy. Before working on the pneumatic system, shut off the system air supply and purge the air lines.
- Isolate the robot from all remote control signals. If maintenance must be done when the power is on, make sure the person inside the work envelope has sole control of the robot. The teach pendant must be held by this person.
- Make sure personnel cannot get trapped between the moving robot and other equipment. Know the path that can be used to escape from a moving robot. Make sure the escape route is never blocked.
- Use blocks, mechanical stops, and pins to prevent hazardous movement by the robot. Make sure that such devices do not create pinch points that could trap personnel.

 WARNING

Do not try to remove any mechanical component from the robot before thoroughly reading and understanding the procedures in the appropriate manual. Doing so can result in serious personal injury and component destruction.

- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.
- When replacing or installing components, make sure dirt and debris do not enter the system.
- Use only specified parts for replacement. To avoid fires and damage to parts in the controller, never use nonspecified fuses.
- Before restarting a robot, make sure no one is inside the work envelope; be sure that the robot and all external devices are operating normally.

KEEPING MACHINE TOOLS AND EXTERNAL DEVICES SAFE

Certain programming and mechanical measures are useful in keeping the machine tools and other external devices safe. Some of these measures are outlined below. Make sure you know all associated measures for safe use of such devices.

Programming Safety Precautions

Implement the following programming safety measures to prevent damage to machine tools and other external devices.

- Back-check limit switches in the workcell to make sure they do not fail.
- Implement “failure routines” in programs that will provide appropriate robot actions if an external device or another robot in the workcell fails.
- Use *handshaking* protocol to synchronize robot and external device operations.
- Program the robot to check the condition of all external devices during an operating cycle.

Mechanical Safety Precautions

Implement the following mechanical safety measures to prevent damage to machine tools and other external devices.

- Make sure the workcell is clean and free of oil, water, and debris.
- Use DCS (Dual Check Safety), software limits, limit switches, and mechanical hardstops to prevent undesired movement of the robot into the work area of machine tools and external devices.

KEEPING THE ROBOT SAFE

Observe the following operating and programming guidelines to prevent damage to the robot.

Operating Safety Precautions

The following measures are designed to prevent damage to the robot during operation.

- Use a low override speed to increase your control over the robot when jogging the robot.
- Visualize the movement the robot will make before you press the jog keys on the teach pendant.
- Make sure the work envelope is clean and free of oil, water, or debris.
- Use circuit breakers to guard against electrical overload.

Programming Safety Precautions

The following safety measures are designed to prevent damage to the robot during programming:

- Establish *interference zones* to prevent collisions when two or more robots share a work area.
- Make sure that the program ends with the robot near or at the home position.
- Be aware of signals or other operations that could trigger operation of tooling resulting in personal injury or equipment damage.
- In dispensing applications, be aware of all safety guidelines with respect to the dispensing materials.

NOTE: Any deviation from the methods and safety practices described in this manual must conform to the approved standards of your company. If you have questions, see your supervisor.

ADDITIONAL SAFETY CONSIDERATIONS FOR PAINT ROBOT INSTALLATIONS

Process technicians are sometimes required to enter the paint booth, for example, during daily or routine calibration or while teaching new paths to a robot. Maintenance personnel also must work inside the paint booth periodically.

Whenever personnel are working inside the paint booth, ventilation equipment must be used. Instruction on the proper use of ventilating equipment usually is provided by the paint shop supervisor.

Although paint booth hazards have been minimized, potential dangers still exist. Therefore, today's highly automated paint booth requires that process and maintenance personnel have full awareness of the system and its capabilities. They must understand the interaction that occurs between the vehicle moving along the conveyor and the robot(s), hood/deck and door opening devices, and high-voltage electrostatic tools.



CAUTION

Ensure that all ground cables remain connected. Never operate the paint robot with ground provisions disconnected. Otherwise, you could injure personnel or damage equipment.

Paint robots are operated in three modes:

- Teach or manual mode
- Automatic mode, including automatic and exercise operation
- Diagnostic mode

During both teach and automatic modes, the robots in the paint booth will follow a predetermined pattern of movements. In teach mode, the process technician teaches (programs) paint paths using the teach pendant.

In automatic mode, robot operation is initiated at the System Operator Console (SOC) or Manual Control Panel (MCP), if available, and can be monitored from outside the paint booth. All personnel must remain outside of the booth or in a designated safe area within the booth whenever automatic mode is initiated at the SOC or MCP.

In automatic mode, the robots will execute the path movements they were taught during teach mode, but generally at production speeds.

When process and maintenance personnel run diagnostic routines that require them to remain in the paint booth, they must stay in a designated safe area.

Paint System Safety Features

Process technicians and maintenance personnel must become totally familiar with the equipment and its capabilities. To minimize the risk of injury when working near robots and related equipment, personnel must comply strictly with the procedures in the manuals.

This section provides information about the safety features that are included in the paint system and also explains the way the robot interacts with other equipment in the system.

The paint system includes the following safety features:

- Most paint booths have red warning beacons that illuminate when the robots are armed and ready to paint. Your booth might have other kinds of indicators. Learn what these are.

- Some paint booths have a blue beacon that, when illuminated, indicates that the electrostatic devices are enabled. Your booth might have other kinds of indicators. Learn what these are.
- EMERGENCY STOP buttons are located on the robot controller and teach pendant. Become familiar with the locations of all E–STOP buttons.
- An intrinsically safe teach pendant is used when teaching in hazardous paint atmospheres.
- A DEADMAN switch is located on each teach pendant. When this switch is held in, and the teach pendant is on, power is applied to the robot servo system. If the engaged DEADMAN switch is released or pressed harder during robot operation, power is removed from the servo system, all axis brakes are applied, and the robot comes to an EMERGENCY STOP. Safety interlocks within the system might also E–STOP other robots.

 WARNING

An EMERGENCY STOP will occur if the DEADMAN switch is released on a bypassed robot.

- Overtravel by robot axes is prevented by software limits. All of the major and minor axes are governed by software limits. DCS (Dual Check Safety), limit switches and hardstops also limit travel by the major axes.
- EMERGENCY STOP limit switches and photoelectric eyes might be part of your system. Limit switches, located on the entrance/exit doors of each booth, will EMERGENCY STOP all equipment in the booth if a door is opened while the system is operating in automatic or manual mode. For some systems, signals to these switches are inactive when the switch on the SOC is in teach mode.
- When present, photoelectric eyes are sometimes used to monitor unauthorized intrusion through the entrance/exit silhouette openings.
- System status is monitored by computer. Severe conditions result in automatic system shutdown.

Staying Safe While Operating the Paint Robot

When you work in or near the paint booth, observe the following rules, in addition to all rules for safe operation that apply to all robot systems.

 WARNING

Observe all safety rules and guidelines to avoid injury.

 WARNING

Never bypass, strap, or otherwise deactivate a safety device, such as a limit switch, for any operational convenience. Deactivating a safety device is known to have resulted in serious injury and death.

 WARNING

Enclosures shall not be opened unless the area is known to be nonhazardous or all power has been removed from devices within the enclosure. Power shall not be restored after the enclosure has been opened until all combustible dusts have been removed from the interior of the enclosure and the enclosure purged. Refer to the Purge chapter for the required purge time.

- Know the work area of the entire paint station (workcell).
- Know the work envelope of the robot and hood/deck and door opening devices.
- Be aware of overlapping work envelopes of adjacent robots.
- Know where all red, mushroom-shaped EMERGENCY STOP buttons are located.
- Know the location and status of all switches, sensors, and/or control signals that might cause the robot, conveyor, and opening devices to move.
- Make sure that the work area near the robot is clean and free of water, oil, and debris. Report unsafe conditions to your supervisor.
- Become familiar with the complete task the robot will perform BEFORE starting automatic mode.
- Make sure all personnel are outside the paint booth before you turn on power to the robot servo system.
- Never enter the work envelope or paint booth before you turn off power to the robot servo system.
- Never enter the work envelope during automatic operation unless a safe area has been designated.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Remove all metallic objects, such as rings, watches, and belts, before entering a booth when the electrostatic devices are enabled.
- Stay out of areas where you might get trapped between a moving robot, conveyor, or opening device and another object.
- Be aware of signals and/or operations that could result in the triggering of guns or bells.
- Be aware of all safety precautions when dispensing of paint is required.
- Follow the procedures described in this manual.

Special Precautions for Combustible Dusts (Powder Paint)

When the robot is used in a location where combustible dusts are found, such as the application of powder paint, the following special precautions are required to insure that there are no combustible dusts inside the robot.

- Purge maintenance air should be maintained at all times, even when the robot power is off. This will insure that dust can not enter the robot.
- A purge cycle will not remove accumulated dusts. Therefore, if the robot is exposed to dust when maintenance air is not present, it will be necessary to remove the covers and clean out any accumulated dust. Do not energize the robot until you have performed the following steps.
 1. Before covers are removed, the exterior of the robot should be cleaned to remove accumulated dust.
 2. When cleaning and removing accumulated dust, either on the outside or inside of the robot, be sure to use methods appropriate for the type of dust that exists. Usually lint free rags dampened with water are acceptable. Do not use a vacuum cleaner to remove dust as it can generate static electricity and cause an explosion unless special precautions are taken.
 3. Thoroughly clean the interior of the robot with a lint free rag to remove any accumulated dust.
 4. When the dust has been removed, the covers must be replaced immediately.
 5. Immediately after the covers are replaced, run a complete purge cycle. The robot can now be energized.

Staying Safe While Operating Paint Application Equipment

When you work with paint application equipment, observe the following rules, in addition to all rules for safe operation that apply to all robot systems.



WARNING

When working with electrostatic paint equipment, follow all national and local codes as well as all safety guidelines within your organization. Also reference the following standards: NFPA 33 Standards for Spray Application Using Flammable or Combustible Materials, and NFPA 70 National Electrical Code.

- **Grounding:** All electrically conductive objects in the spray area must be grounded. This includes the spray booth, robots, conveyors, workstations, part carriers, hooks, paint pressure pots, as well as solvent containers. Grounding is defined as the object or objects shall be electrically connected to ground with a resistance of not more than 1 megohms.
- **High Voltage:** High voltage should only be on during actual spray operations. Voltage should be off when the painting process is completed. Never leave high voltage on during a cap cleaning process.
- Avoid any accumulation of combustible vapors or coating matter.
- Follow all manufacturer recommended cleaning procedures.
- Make sure all interlocks are operational.

- No smoking.
- Post all warning signs regarding the electrostatic equipment and operation of electrostatic equipment according to NFPA 33 Standard for Spray Application Using Flammable or Combustible Material.
- Disable all air and paint pressure to bell.
- Verify that the lines are not under pressure.

Staying Safe During Maintenance

When you perform maintenance on the painter system, observe the following rules, and all other maintenance safety rules that apply to all robot installations. Only qualified, trained service or maintenance personnel should perform repair work on a robot.

- Paint robots operate in a potentially explosive environment. Use caution when working with electric tools.
- When a maintenance technician is repairing or adjusting a robot, the work area is under the control of that technician. All personnel not participating in the maintenance must stay out of the area.
- For some maintenance procedures, station a second person at the control panel within reach of the EMERGENCY STOP button. This person must understand the robot and associated potential hazards.
- Be sure all covers and inspection plates are in good repair and in place.
- Always return the robot to the “home” position before you disarm it.
- Never use machine power to aid in removing any component from the robot.
- During robot operations, be aware of the robot’s movements. Excess vibration, unusual sounds, and so forth, can alert you to potential problems.
- Whenever possible, turn off the main electrical disconnect before you clean the robot.
- When using vinyl resin observe the following:
 - Wear eye protection and protective gloves during application and removal.
 - Adequate ventilation is required. Overexposure could cause drowsiness or skin and eye irritation.
 - If there is contact with the skin, wash with water.
 - Follow the Original Equipment Manufacturer’s Material Safety Data Sheets.
- When using paint remover observe the following:
 - Eye protection, protective rubber gloves, boots, and apron are required during booth cleaning.
 - Adequate ventilation is required. Overexposure could cause drowsiness.
 - If there is contact with the skin or eyes, rinse with water for at least 15 minutes. Then seek medical attention as soon as possible.
 - Follow the Original Equipment Manufacturer’s Material Safety Data Sheets.

SAFETY PRECAUTIONS

This chapter describes the precautions which must be followed to ensure the safe use of the robot. Before using the robot, be sure to read this chapter thoroughly.

For detailed functions of the robot operation, read the relevant operator's manual to understand fully its specification.

For the safety of the operator and the system, follow all safety precautions when operating a robot and its peripheral equipment installed in a work cell.

In addition, refer to the "FANUC Robot SAFETY HANDBOOK (B-80687EN)".

1 DEFINITION OF USER

The user can be defined as follows.

Operator:

- Turns ON/OFF power to the robot
- Starts the robot program from the operator's panel

Programmer:

- Operates the robot
- Teaches the robot inside the safety fence

Maintenance engineer:

- Operates the robot
- Teaches the robot inside the safety fence
- Performs maintenance (repair, adjustment, replacement)

- Operator is not allowed to work in the safety fence.
- Programmers and maintenance engineers are allowed to work in the safety fence. The work inside the safety fence includes lifting, setting, teaching, adjustment, maintenance, etc.
- To work inside the safety fence, the person must receive a professional training for the robot.

During the operation, programming, and maintenance of your robotic system, the programmer, operator, and maintenance engineer should take additional care of their safety by wearing the following safety items.

- Adequate clothes for the operation
- Safety shoes
- A helmet

2 DEFINITION OF SAFETY NOTATIONS

To ensure the safety of users and prevent damage to the machine, this manual indicates each precaution on safety with "WARNING" or "CAUTION" according to its severity. Supplementary information is indicated by "NOTE". Read the contents of each "WARNING", "CAUTION" and "NOTE" before using the robot.

Symbol	Definitions
 WARNING	Used if hazard resulting in the death or serious injury of the user will be expected to occur if he or she fails to follow the approved procedure.
 CAUTION	Used if a hazard resulting in the minor or moderate injury of the user, or equipment damage may be expected to occur if he or she fails to follow the approved procedure.
NOTE	Used if a supplementary explanation not related to any of WARNING and CAUTION is to be indicated.

- Check this manual thoroughly, and keep it handy for the future reference.

3 SAFETY OF THE USER

User safety is the primary safety consideration. Because it is very dangerous to enter the operating space of the robot during automatic operation, adequate safety precautions must be observed. The following lists the general safety precautions. Careful consideration must be made to ensure user safety.

- (1) Have the robot system users attend the training courses held by FANUC.

FANUC provides various training courses. Contact our sales office for details.

- (2) Even when the robot is stationary, it is possible that the robot is still in a ready to move state, and is waiting for a signal. In this state, the robot is regarded as still in motion. To ensure user safety, provide the system with an alarm to indicate visually or aurally that the robot is in motion.
- (3) Install a safety fence with a gate so that no user can enter the work area without passing through the gate. Install an interlocking device, a safety plug, and so forth in the safety gate so that the robot is stopped as the safety gate is opened.

The controller is designed to receive this interlocking signal of the door switch. When the gate is opened and this signal received, the controller stops the robot (Please refer to "STOP TYPE OF ROBOT" in "SAFETY PRECAUTIONS" for detail of stop type). For connection, see Fig. 3 (b).

- (4) Provide the peripheral equipment with appropriate earth (Class A, Class B, Class C, and Class D).
- (5) Try to install the peripheral equipment outside the robot operating space.
- (6) Draw an outline on the floor, clearly indicating the range of the robot operating space, including the tools such as a hand.
- (7) Install a mat switch or photoelectric switch on the floor with an interlock to a visual or aural alarm that stops the robot when a user enters the work area.
- (8) If necessary, install a safety lock so that no one except the user in charge can turn on the power of the robot.

The circuit breaker installed in the controller is designed to disable anyone from turning it on when it is locked with a padlock.

- (9) When adjusting each peripheral equipment independently, be sure to turn off the power of the robot.
- (10) Operators should be ungloved while manipulating the operator panel or teach pendant. Operation with gloved fingers could cause an operation error.
- (11) Programs, system variables, and other information can be saved on memory card or USB memories. Be sure to save the data periodically in case the data is lost in an accident. (refer to Controller OPERATOR'S MANUAL.)
- (12) The robot should be transported and installed by accurately following the procedures recommended by FANUC. Wrong transportation or installation may cause the robot to fall, resulting in severe injury to workers.
- (13) In the first operation of the robot after installation, the operation should be restricted to low speeds. Then, the speed should be gradually increased to check the operation of the robot.
- (14) Before the robot is started, it should be checked that no one is inside the safety fence. At the same time, a check must be made to ensure that there is no risk of hazardous situations. If detected, such a situation should be eliminated before the operation.
- (15) When the robot is used, the following precautions should be taken. Otherwise, the robot and peripheral equipment can be adversely affected, or workers can be severely injured.
 - Avoid using the robot in a flammable environment.
 - Avoid using the robot in an explosive environment.
 - Avoid using the robot in an environment full of radiation.
 - Avoid using the robot under water or at high humidity.
 - Avoid using the robot to carry a person or animal.
 - Avoid using the robot as a stepladder. (Never climb up on or hang from the robot.)
 - Outdoor
- (16) When connecting the peripheral equipment related to stop (safety fence etc.) and each signal (external emergency, fence etc.) of robot, be sure to confirm the stop movement and do not take the wrong connection.
- (17) When preparing footstep, please consider security for installation and maintenance work in high place according to Fig. 3 (c). Please consider footstep and safety belt mounting position.

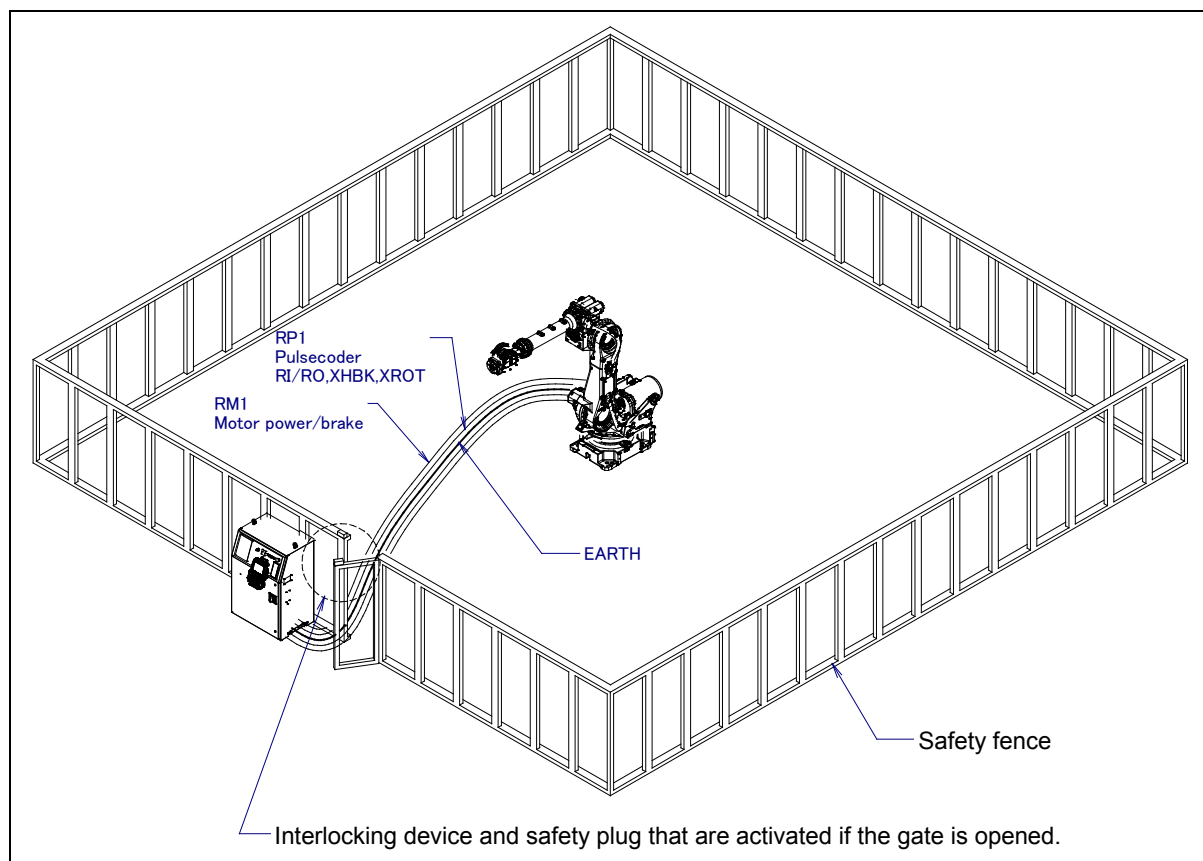


Fig. 3 (a) Safety fence and safety gate

**WARNING**

When you close a fence, please confirm that there is not a person from all directions of the robot.

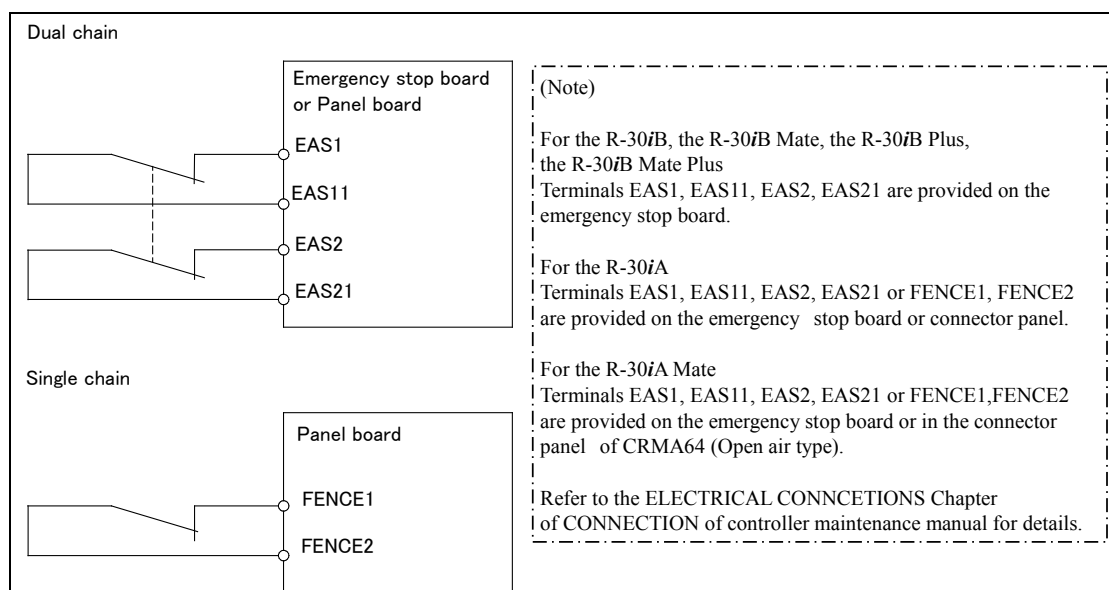


Fig. 3 (b) Connection diagram for the signal of safety fence

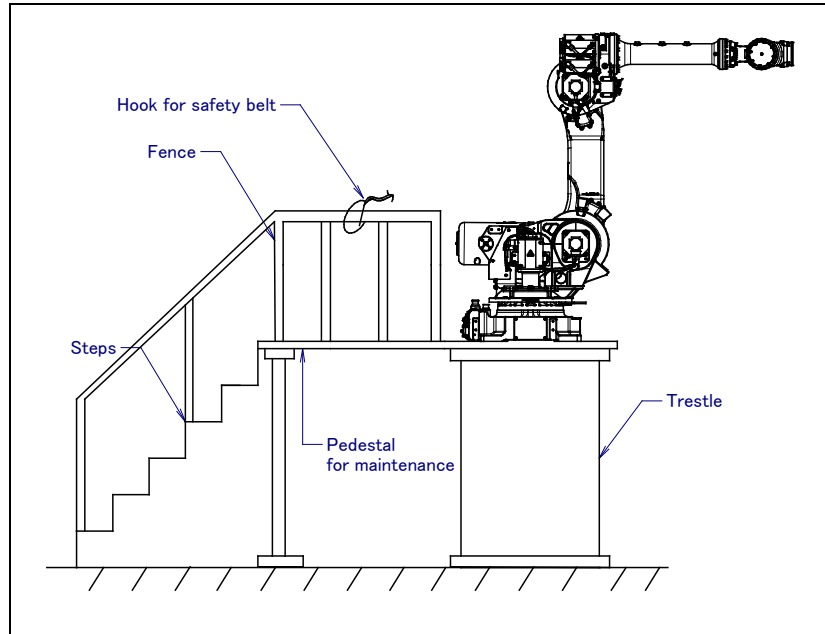


Fig. 3 (c) Pedestal for maintenance

3.1 SAFETY OF THE OPERATOR

An operator refers to a person who turns on and off the robot system and starts a robot program from, for example, the operator panel during daily operation.

Operators cannot work inside of the safety fence.

- (1) If the robot does not need to be operated, turn off the robot controller power or press the EMERGENCY STOP button during working.
- (2) Operate the robot system outside the operating space of the robot.
- (3) Install a safety fence or safety door to avoid the accidental entry of a person other than an operator in charge or keep operator out from the hazardous place.
- (4) Install the EMERGENCY STOP button within the operator's reach.

The robot controller is designed to be connected to an external EMERGENCY STOP button. With this connection, the controller stops the robot operation (Please refer to "STOP TYPE OF ROBOT" in "SAFETY PRECAUTIONS" for detail of stop type) when the external EMERGENCY STOP button is pressed. See the diagram below for connection.

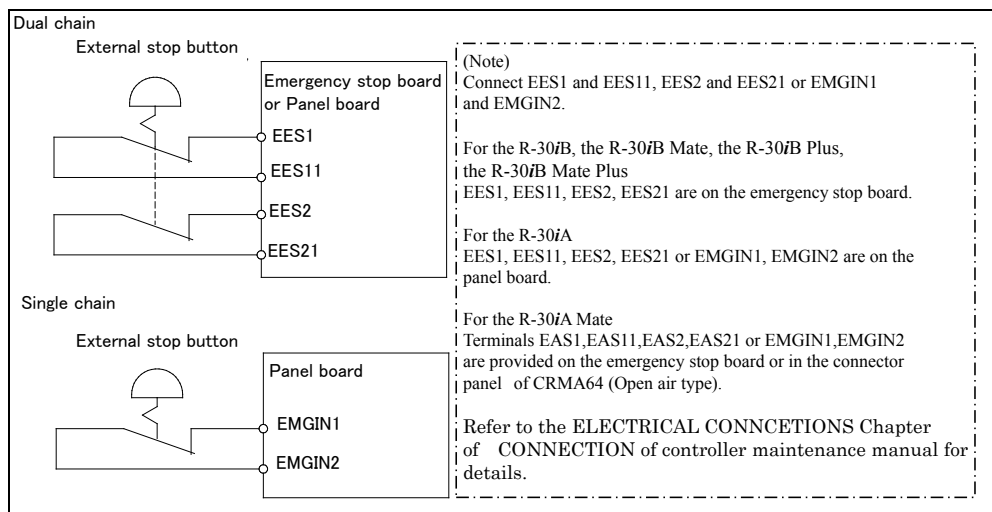


Fig. 3.1 Connection diagram for external emergency stop button

3.2 SAFETY OF THE PROGRAMMER

While teaching the robot, the operator may need to enter the robot operation area. The programmer must ensure the safety especially.

- (1) Unless it is specifically necessary to enter the robot operating space, carry out all tasks outside the operating space.
- (2) Before teaching the robot, check that the robot and its peripheral equipment are all in the normal operating condition.
- (3) If it is inevitable to enter the robot operating space to teach the robot, check the locations, settings, and other conditions of the safety devices (such as the EMERGENCY STOP button, the DEADMAN switch on the teach pendant) before entering the area.
- (4) The programmer must be extremely careful not to let anyone else enter the robot operating space.
- (5) Programming should be done outside the area of the safety fence as far as possible. If programming needs to be done inside the safety fence, the programmer should take the following precautions:
 - Before entering the area of the safety fence, ensure that there is no risk of dangerous situations in the area.
 - Be prepared to press the emergency stop button whenever necessary.
 - Robot motions should be made at low speeds.
 - Before starting programming, check the whole robot system status to ensure that no remote instruction to the peripheral equipment or motion would be dangerous to the user.

Our operator panel is provided with an emergency stop button and a key switch (mode switch) for selecting the automatic operation mode (AUTO) and the teach modes (T1 and T2). Before entering the inside of the safety fence for the purpose of teaching, set the switch to a teach mode, remove the key from the mode switch to prevent other people from changing the operation mode carelessly, then open the safety gate. If the safety gate is opened with the automatic operation mode set, the robot stops (Please refer to "**STOP TYPE OF ROBOT**" in SAFETY PRECAUTIONS for detail of stop type). After the switch is set to a teach mode, the safety gate is disabled. The programmer should understand that the safety gate is disabled and is responsible for keeping other people from entering the inside of the safety fence.

Teach pendant is provided with a switch to enable/disable robot operation from teach pendant and DEADMAN switch as well as emergency stop button. These button and switch function as follows:

- (1) Emergency stop button: Causes the stop of the robot (Please refer to "STOP TYPE OF ROBOT" in SAFETY PRECAUTIONS for detail of stop type) when pressed.
- (2) DEADMAN switch: Functions are different depending on the teach pendant enable/disable switch setting status.
 - (a) **Enable:** Servo power is turned off and robot stops when the operator releases the DEADMAN switch or when the operator presses the switch strongly.
 - (b) **Disable:** The DEADMAN switch is disabled.

(Note) The DEADMAN switch is provided to stop the robot when the operator releases the teach pendant or presses the pendant strongly in case of emergency. The R-30iB Plus/R-30iB Mate Plus /R-30iB/R-30iB Mate/R-30iA/R-30iA Mate employs a 3-position DEADMAN switch, which allows the robot to operate when the 3-position DEADMAN switch is pressed to its intermediate point. When the operator releases the DEADMAN switch or presses the switch strongly, the robot stops immediately.

The operator's intention of starting teaching is determined by the controller through the dual operation of setting the teach pendant enable/disable switch to the enable position and pressing the DEADMAN switch. The operator should make sure that the robot could operate in such conditions and be responsible in carrying out tasks safely.

Based on the risk assessment by FANUC, number of operation of DEADMAN switch should not exceed about 10000 times per year.

The teach pendant, operator panel, and peripheral device interface send each robot start signal. However the validity of each signal changes as follows depending on the mode switch and the DEADMAN switch of the operator panel, the teach pendant enable switch and the remote condition on the software.

**For the R-30iB Plus/R-30iB Mate Plus/R-30iB/R-30iB Mate/R-30iA Controller
or CE or RIA specification of the R-30iA Mate Controller**

Mode	Teach pendant enable switch	Software remote condition	Teach pendant	Operator panel	Peripheral device
AUTO mode	On	Local	Not allowed	Not allowed	Not allowed
		Remote	Not allowed	Not allowed	Not allowed
	Off	Local	Not allowed	Allowed to start	Not allowed
		Remote	Not allowed	Not allowed	Allowed to start
T1, T2 mode	On	Local	Allowed to start	Not allowed	Not allowed
		Remote	Allowed to start	Not allowed	Not allowed
	Off	Local	Not allowed	Not allowed	Not allowed
		Remote	Not allowed	Not allowed	Not allowed

T1,T2 mode: DEADMAN switch is effective.

For the standard specification of R-30iA Mate Controller

Teach pendant enable switch	Software remote condition	Teach pendant	Peripheral device
On	Ignored	Allowed to start	Not allowed
Off	Local	Not allowed	Not allowed
	Remote	Not allowed	Allowed to start

- (6) (Only when R-30iB Plus/R-30iB Mate Plus/R-30iB/R-30iB Mate /R-30iA Controller or CE or RIA specification of R-30iA Mate controller is selected.) To start the system using the operator panel, make certain that nobody is in the robot operating space and that there are no abnormal conditions in the robot operating space.
- (7) When a program is completed, be sure to carry out the test operation according to the following procedure.
 - (a) Run the program for at least one operation cycle in the single step mode at low speed.
 - (b) Run the program for at least one operation cycle in the continuous operation mode at low speed.
 - (c) Run the program for one operation cycle in the continuous operation mode at the intermediate speed and check that no abnormalities occur due to a delay in timing.
 - (d) Run the program for one operation cycle in the continuous operation mode at the normal operating speed, and check that the system operates automatically without trouble.
 - (e) After checking the completeness of the program through the test operation above, execute it in the automatic operation mode.
- (8) While operating the system in the automatic operation mode, the teach pendant operator must leave the safety fence.

3.3 SAFETY OF THE MAINTENANCE ENGINEER

For the safety of maintenance engineer personnel, pay utmost attention to the following.

- (1) During operation, never enter the robot operating space.
- (2) A hazardous situation may arise when the robot or the system, are kept with their power-on during maintenance operations. Therefore, for any maintenance operation, the robot and the system should be put into the power-off state. If necessary, a lock should be in place in order to prevent any other person from turning on the robot and/or the system. In case maintenance needs to be executed in the power-on state, the emergency stop button must be pressed.
- (3) If it becomes necessary to enter the robot operating space while the power is on, press the emergency stop button on the operator box or operator panel, or the teach pendant before entering the range. The maintenance worker must indicate that maintenance work is in progress and be careful not to allow other people to operate the robot carelessly.
- (4) When entering the area enclosed by the safety fence, the worker must check the whole robot system in order to make sure no dangerous situations exist. In case the worker needs to enter the safety area whilst a dangerous situation exists, extreme care must be taken, and whole robot system status must be carefully monitored.
- (5) Before the maintenance of the pneumatic system is started, the supply pressure should be shut off and the pressure in the piping should be reduced to zero.
- (6) Before the start of maintenance work, check that the robot and its peripheral equipment are all in the normal operating condition.
- (7) Do not operate the robot in the automatic operation while anybody is in the robot operating space.
- (8) When you maintain the robot alongside a wall or instrument, or when multiple users are working nearby, make certain that their escape path is not obstructed.
- (9) When a tool is mounted on the robot, or when any movable device other than the robot is installed, such as belt conveyor, pay careful attention to its motion.
- (10) If necessary, have a user who is familiar with the robot system stand beside the operator panel and observe the work being performed. If any danger arises, the user should be ready to press the EMERGENCY STOP button at any time.
- (11) When replacing a part, please contact your local FANUC representative. If a wrong procedure is followed, an accident may occur, causing damage to the robot and injury to the user.
- (12) When replacing or reinstalling components, take care to prevent foreign material from entering the system.
- (13) When handling each unit or printed circuit board in the controller during inspection, turn off the circuit breaker to protect against electric shock.
If there are two cabinets, turn off the both circuit breaker.
- (14) A part should be replaced with a part recommended by FANUC. If other parts are used, malfunction or damage would occur. Especially, a fuse that is not recommended by FANUC should not be used. Such a fuse may cause a fire.
- (15) When restarting the robot system after completing maintenance work, make sure in advance that there is no person in the operating space and that the robot and the peripheral equipment are not abnormal.
- (16) When a motor or brake is removed, the robot arm should be supported with a crane or other equipment beforehand so that the arm would not fall during the removal.
- (17) Whenever grease is spilled on the floor, it should be removed as quickly as possible to prevent dangerous falls.
- (18) The following parts are heated. If a maintenance user needs to touch such a part in the heated state, the user should wear heat-resistant gloves or use other protective tools.
 - Servo motor
 - Inside the controller
 - Reducer
 - Gearbox
 - Wrist unit

- (19) Maintenance should be done under suitable light. Care must be taken that the light would not cause any danger.
- (20) When a motor, reducer, or other heavy load is handled, a crane or other equipment should be used to protect maintenance workers from excessive load. Otherwise, the maintenance workers would be severely injured.
- (21) The robot should not be stepped on or climbed up during maintenance. If it is attempted, the robot would be adversely affected. In addition, a misstep can cause injury to the worker.
- (22) When performing maintenance work in high place, secure a footstep and wear safety belt.
- (23) After the maintenance is completed, spilled oil or water and metal chips should be removed from the floor around the robot and within the safety fence.
- (24) When a part is replaced, all bolts and other related components should put back into their original places. A careful check must be given to ensure that no components are missing or left not mounted.
- (25) In case robot motion is required during maintenance, the following precautions should be taken :
 - Foresee an escape route. And during the maintenance motion itself, monitor continuously the whole robot system so that your escape route will not become blocked by the robot, or by peripheral equipment.
 - Always pay attention to potentially dangerous situations, and be prepared to press the emergency stop button whenever necessary.
- (26) The robot should be periodically inspected. (Refer to the robot mechanical manual and controller maintenance manual.) A failure to do the periodical inspection can adversely affect the performance or service life of the robot and may cause an accident
- (27) After a part is replaced, a test execution should be given for the robot according to a predetermined method. (See TESTING section of "Controller operator's manual".) During the test execution, the maintenance worker should work outside the safety fence.

4 SAFETY OF THE TOOLS AND PERIPHERAL EQUIPMENT

4.1 PRECAUTIONS IN PROGRAMMING

- (1) Use a limit switch or other sensor to detect a dangerous condition and, if necessary, design the program to stop the robot when the sensor signal is received.
- (2) Design the program to stop the robot when an abnormality occurs in any other robots or peripheral equipment, even though the robot itself is normal.
- (3) For a system in which the robot and its peripheral equipment are in synchronous motion, particular care must be taken in programming so that they do not interfere with each other.
- (4) Provide a suitable interface between the robot and its peripheral equipment so that the robot can detect the states of all devices in the system and can be stopped according to the states.

4.2 PRECAUTIONS FOR MECHANISM

- (1) Keep the component cells of the robot system clean, operate the robot where insulated from the influence of oil, water, and dust.
- (2) Don't use unconfirmed liquid for cutting fluid and cleaning fluid.
- (3) Adopt limit switches or mechanical stoppers to limit the robot motion, and avoid the robot from collisions against peripheral equipment or tools.
- (4) Observe the following precautions about the mechanical unit cables. Failure to follow precautions may cause problems.
 - Use mechanical unit cable that have required user interface.
 - Do not add user cable or hose to inside of the mechanical unit.
 - Please do not obstruct the movement of the mechanical unit when cables are added to outside of mechanical unit.
 - In the case of the model that a cable is exposed, please do not perform remodeling (Adding a protective cover and fix an outside cable more) obstructing the behavior of the outcrop of the cable.
 - When installing user peripheral equipment on the robot mechanical unit, please pay attention that the device does not interfere with the robot itself.
- (5) The frequent power-off stop for the robot during operation causes the trouble of the robot. Please avoid the system construction that power-off stop would be operated routinely. (Refer to bad case example.) Please perform power-off stop after reducing the speed of the robot and stopping it by hold stop or cycle stop when it is not urgent. (Please refer to "STOP TYPE OF ROBOT" in "SAFETY PRECAUTIONS" for detail of stop type.)

(Bad case example)

 - Whenever poor product is generated, a line stops by emergency stop and power-off of the robot is incurred.
 - When alteration is necessary, safety switch is operated by opening safety fence and power-off stop is incurred for the robot during operation.
 - An operator pushes the emergency stop button frequently, and a line stops.
 - An area sensor or a mat switch connected to safety signal operates routinely and power-off stop is incurred for the robot.
 - Power-off stop is regularly incurred due to an inappropriate setting for Dual Check Safety (DCS).
- (6) Power-off stop of Robot is executed when collision detection alarm (SRVO-050) etc. occurs. Please try to avoid unnecessary power-off stops. It may cause the trouble of the robot, too. So remove the causes of the alarm.

5 SAFETY OF THE ROBOT MECHANICAL UNIT

5.1 PRECAUTIONS IN OPERATION

- (1) When operating the robot in the jog mode, set it at an appropriate speed so that the operator can manage the robot in any eventuality.
- (2) Before pressing the jog key, be sure you know in advance what motion the robot will perform in the jog mode.

5.2 PRECAUTIONS IN PROGRAMMING

- (1) When the operating spaces of robots overlap, make certain that the motions of the robots do not interfere with each other.
- (2) Be sure to specify the predetermined work origin in a motion program for the robot and program the motion so that it starts from the origin and terminates at the origin. Make it possible for the operator to easily distinguish at a glance that the robot motion has terminated.

5.3 PRECAUTIONS FOR MECHANISMS

Keep the robot operation area clean, and operate the robot in an environment free of grease, water, and dust.

5.4 PROCEDURE TO MOVE ARM WITHOUT DRIVE POWER IN EMERGENCY OR ABNORMAL SITUATIONS

For emergency or abnormal situations (e.g. persons trapped in or pinched by the robot), brake release unit can be used to move the robot axes without drive power.

Please refer to controller maintenance manual and mechanical unit operator's manual for using method of brake release unit and method of supporting robot.

6 SAFETY OF THE END EFFECTOR

6.1 PRECAUTIONS IN PROGRAMMING

- (1) To control the pneumatic, hydraulic and electric actuators, carefully consider the necessary time delay after issuing each control command up to actual motion and ensure safe control.
- (2) Provide the end effector with a limit switch, and control the robot system by monitoring the state of the end effector.

7 STOP TYPE OF ROBOT (R-30iA, R-30iA Mate)

The following three robot stop types exist:

Power-Off Stop (Category 0 following IEC 60204-1)

Servo power is turned off and the robot stops immediately. Servo power is turned off when the robot is moving, and the motion path of the deceleration is uncontrolled.

The following processing is performed at Power-Off stop.

- An alarm is generated and servo power is turned off.
- The robot operation is stopped immediately. Execution of the program is paused.

Frequent Power-Off stop of the robot during operation can cause mechanical problems of the robot.

Avoid system designs that require routine or frequent Power-Off stop conditions.

Controlled stop (Category 1 following IEC 60204-1)

The robot is decelerated until it stops, and servo power is turned off.

The following processing is performed at Controlled stop.

- The alarm "SRVO-199 Controlled stop" occurs along with a decelerated stop. Execution of the program is paused.
- An alarm is generated and servo power is turned off.

Hold (Category 2 following IEC 60204-1)

The robot is decelerated until it stops, and servo power remains on.

The following processing is performed at Hold.

- The robot operation is decelerated until it stops. Execution of the program is paused.

WARNING

- 1 The stopping distance and time of Controlled stop is longer than those of Power-Off stop. A risk assessment for the whole robot system which takes into consideration the increased stopping distance and stopping time is necessary when Controlled stop is used. Please refer to the operator's manual of a particular robot model for the data of stopping distance and time.
- 2 In multi arm system, the longest stopping distance and time of Controlled Stop among each robot are adopted as those for the system. A risk assessment for the whole robot system which takes into consideration a possibility that the stopping distance and time increase, is necessary on the multi arm system.
- 3 In the system which has extended axis, the longer stopping distance and time of Controlled Stop among robot and extended axis are adopted as those for the system. A risk assessment for the whole robot system which takes into consideration a possibility that the stopping distance and time increase, is necessary on the system which has extended axis. Please refer to the extended axis setup procedure of the controller operator's manual for considering the stopping distance and time of the extended axis.
- 4 In case of Controlled stop, motor power shutdown is delayed for a maximum of 2 seconds. In this case, a risk assessment for the whole robot system is necessary, including the 2 seconds delay.

When the emergency stop button is pressed or the FENCE is open, the stop type of robot is Power-Off stop or Controlled stop. The configuration of stop type for each situation is called *stop pattern*. The stop pattern is different according to the controller type or option configuration.

There are the following 3 Stop patterns.

Stop pattern	Mode	Emergency stop button	External Emergency stop	FENCE open	SVOFF input	Servo disconnect
A	AUTO	P-Stop	P-Stop	C-Stop	C-Stop	P-Stop
	T1	P-Stop	P-Stop	-	C-Stop	P-Stop
	T2	P-Stop	P-Stop	-	C-Stop	P-Stop
B	AUTO	P-Stop	P-Stop	P-Stop	P-Stop	P-Stop
	T1	P-Stop	P-Stop	-	P-Stop	P-Stop
	T2	P-Stop	P-Stop	-	P-Stop	P-Stop
C	AUTO	C-Stop	C-Stop	C-Stop	C-Stop	C-Stop
	T1	P-Stop	P-Stop	-	C-Stop	P-Stop
	T2	P-Stop	P-Stop	-	C-Stop	P-Stop

P-Stop: Power-Off stop

C-Stop: Controlled stop

-: Disable

The following table indicates the Stop pattern according to the controller type or option configuration.

Option	R-30iA				R-30iA Mate		
	Standard (Single)	Standard (Dual)	RIA type	CE type	Standard	RIA type	CE type
Standard	B (*)	A	A	A	A (**)	A	A
Stop type set (Stop pattern C) (A05B-2500-J570)	N/A	N/A	C	C	N/A	C	C

(*) R-30iA standard (single) does not have servo disconnect.

(**) R-30iA Mate Standard does not have servo disconnect, and the stop type of SVOFF input is Power-Off stop.

The stop pattern of the controller is displayed in "Stop pattern" line in software version screen. Please refer to "Software version" in operator's manual of controller for the detail of software version screen.

"Controlled stop by E-Stop" option

When "Stop type set (Stop pattern C) (A05B-2500-J570) option is specified, the stop type of the following alarms becomes Controlled stop but only in AUTO mode. In T1 or T2 mode, the stop type is Power-Off stop which is the normal operation of the system.

Alarm	Condition
SRVO-001 Operator panel E-stop	Operator panel emergency stop is pressed.
SRVO-002 Teach pendant E-stop	Teach pendant emergency stop is pressed.
SRVO-007 External emergency stops	External emergency stop input (EES1-EES11, EES2-EES21) is open. (R-30iA controller)
SRVO-194 Servo disconnect	Servo disconnect input (SD4-SD41, SD5-SD51) is open. (R-30iA controller)
SRVO-218 Ext.E-stop/Servo Disconnect	External emergency stop input (EES1-EES11, EES2-EES21) is open. (R-30iA Mate controller)
SRVO-408 DCS SSO Ext Emergency Stop	In DCS Safe I/O connect function, SSO[3] is OFF.
SRVO-409 DCS SSO Servo Disconnect	In DCS Safe I/O connect function, SSO[4] is OFF.

Controlled stop is different from Power-Off stop as follows:

- In Controlled stop, the robot is stopped on the program path. This function is effective for a system where the robot can interfere with other devices if it deviates from the program path.
- In Controlled stop, physical impact is less than Power-Off stop. This function is effective for systems where the physical impact to the mechanical unit or EOAT (End Of Arm Tool) should be minimized.
- The stopping distance and stopping time of Controlled stop is longer than the stopping distance and stopping time of Power-Off stop, depending on the robot model and axis. Please refer to the operator's manual of a particular robot model for the data of stopping distance and stopping time.

For the R-30iA or R-30iA Mate, this function is available only in CE or RIA type hardware.

When this option is loaded, this function cannot be disabled.

The stop type of DCS Position and Speed Check functions is not affected by the loading of this option.



WARNING

The stopping distance and stopping time of Controlled stop are longer than the stopping distance and stopping time of Power-Off stop. A risk assessment for the whole robot system, which takes into consideration the increased stopping distance and stopping time, is necessary when this option is loaded.

8 STOP TYPE OF ROBOT (R-30iB, R-30iB Mate)

There are following four types of Stopping Robot.

Power-Off Stop (Category 0 following IEC 60204-1)

Servo power is turned off, and the robot stops immediately. Servo power is turned off when the robot is moving, and the motion path of the deceleration is uncontrolled.

“**Power-Off stop**” performs following processing.

- An alarm is generated, and then the servo power turns off. Instantly the robot stops.
- Execution of the program is paused.

Frequent Power-Off stop of the robot during operation can cause mechanical problems of the robot.

Avoid system designs that require routine or frequent Power-Off stop conditions.

Controlled stop (Category 1 following IEC 60204-1)

The robot is decelerated until it stops, and servo power is turned off.

“**Controlled stop**” performs following processing.

- The alarm "**SRVO-199 Controlled stop**" occurs along with a decelerated stop. The program execution is paused.
- An alarm is generated, and then the servo power turns off.

Smooth stop (Category 1 following IEC 60204-1)

The robot is decelerated until it stops, and servo power is turned off.

“**Smooth stop**” performs following processing.

- The alarm "**SRVO-289 Smooth Stop**" occurs along with a decelerated stop. The program execution is paused.
- An alarm is generated, and then the servo power turns off.

- In Smooth stop, the robot decelerates until it stops with the deceleration time shorter than Controlled stop.

Hold (Category 2 following IEC 60204-1)

The robot is decelerated until it stops, and servo power remains on.

“Hold” performs following processing.

- The robot operation is decelerated until it stops. Execution of the program is paused.

WARNING

- 5 The stopping distance and time of Controlled stop and Smooth stop are longer than those of Power-Off stop. A risk assessment for the whole robot system which takes into consideration the increased stopping distance and stopping time is necessary when Controlled stop or Smooth Stop is used. Please refer to the operator's manual of a particular robot model for the data of stopping distance and time.
- 6 In multi arm system, the longest stopping distance and time of Controlled Stop or Smooth Stop among each robot are adopted as those for the system. A risk assessment for the whole robot system which takes into consideration a possibility that the stopping distance and time increase, is necessary on the multi arm system.
- 7 In the system which has extended axis, the longer stopping distance and time of Controlled Stop or Smooth Stop among robot and extended axis are adopted as those for the system. A risk assessment for the whole robot system which takes into consideration a possibility that the stopping distance and time increase, is necessary on the system which has extended axis. Please refer to the extended axis setup procedure of the controller operator's manual for considering the stopping distance and time of the extended axis.
- 8 When Smooth stop occurs during deceleration by Controlled stop, the stop type of robot is changed to Power-Off Stop.
When Smooth stop occurs during deceleration by Hold, the stop type of robot is changed to Power-Off Stop.
- 9 In case of Controlled stop or Smooth Stop, motor power shutdown is delayed for a maximum of 2 seconds. In this case, a risk assessment for the whole robot system is necessary, including the 2 seconds delay.

When the emergency stop button is pressed or the FENCE is open, the stop type of robot is Power-Off stop, Controlled stop, or Smooth stop. The configuration of stop type for each situation is called *stop pattern*. The stop pattern is different according to the option configuration.

There are the following 3 Stop patterns.

Stop pattern	Mode	Emergency stop button	External Emergency stop	FENCE open	SVOFF input	Deadman switch (*)
A	AUTO	P-Stop	P-Stop	C-Stop	C-Stop	-
	T1	P-Stop	P-Stop	-	C-Stop	P-Stop
	T2	P-Stop	P-Stop	-	C-Stop	P-Stop
C	AUTO	C-Stop	C-Stop	C-Stop	C-Stop	-
	T1	P-Stop	P-Stop	-	C-Stop	P-Stop
	T2	P-Stop	P-Stop	-	C-Stop	P-Stop
D	AUTO	S-Stop	S-Stop	C-Stop	C-Stop	-
	T1	S-Stop	S-Stop	-	C-Stop	S-Stop
	T2	S-Stop	S-Stop	-	C-Stop	S-Stop

P-Stop: Power-Off stop

C-Stop: Controlled stop

S-Stop: Smooth stop

-: Disable

(*) The stop pattern of NTED input is same as Deadman switch.

The following table indicates the Stop pattern according to the controller type or option configuration.

Option	R-30iB/ R-30iB Mate
Standard	A(**)
Controlled stop by E-Stop (A05B-2600-J570)	C(**)
Smooth E-Stop (A05B-2600-J651)	D(**)

(**)R-30iB Mate does not have SVOFF input.

The stop pattern of the controller is displayed in "Stop pattern" line in software version screen. Please refer to "Software version" in operator's manual of controller for the detail of software version screen.

"Controlled stop by E-Stop" option

When "Controlled stop by E-Stop" (A05B-2600-J570) option is specified, the stop type of the following alarms become Controlled stop but only in AUTO mode. In T1 or T2 mode, the stop type is Power-Off stop which is the normal operation of the system.

Alarm	Condition
SRVO-001 Operator panel E-stop	Operator panel emergency stop is pressed.
SRVO-002 Teach pendant E-stop	Teach pendant emergency stop is pressed.
SRVO-007 External emergency stops	External emergency stop input (EES1-EES11, EES2-EES21) is open.
SRVO-408 DCS SSO Ext Emergency Stop	In DCS Safe I/O connect function, SSO[3] is OFF.
SRVO-409 DCS SSO Servo Disconnect	In DCS Safe I/O connect function, SSO[4] is OFF.

Controlled stop is different from **Power-Off stop** as follows:

- In Controlled stop, the robot is stopped on the program path. This function is effective for a system where the robot can interfere with other devices if it deviates from the program path.
- In Controlled stop, physical impact is less than Power-Off stop. This function is effective for systems where the physical impact to the mechanical unit or EOAT (End Of Arm Tool) should be minimized.
- The stopping distance and time of Controlled stop is longer than those of Power-Off stop, depending on the robot model and axis.

When this option is loaded, this function cannot be disabled.

The stop type of DCS Position and Speed Check functions is not affected by the loading of this option.

**WARNING**

The stopping distance and time of Controlled stop are longer than those of Power-Off stop. A risk assessment for the whole robot system which takes into consideration the increased stopping distance and stopping time, is necessary when this option is loaded.

"Smooth E-Stop Function" option

When "Smooth E-Stop Function" (A05B-2600-J651) option is specified, the stop type of the following alarms becomes Smooth stop in all operation modes (AUTO, T1 and T2 mode).

Alarm	Condition
SRVO-001 Operator panel E-stop	Operator panel emergency stop is pressed.
SRVO-002 Teach pendant E-stop	Teach pendant emergency stop is pressed.
SRVO-003 Deadman switch released	Both deadman switches on Teach pendant are released.
SRVO-007 External emergency stops	External emergency stop input (EES1-EES11, EES2-EES21) is open.
SRVO-037 IMSTP input (Group: %d)	IMSTP input (*IMSTP signal for a peripheral device interface) is OFF.
SRVO-232 NTED input	NTED input (NTED1-NTED11, NTED2-NTED21) is open.
SRVO-408 DCS SSO Ext Emergency Stop	In DCS Safe I/O connect function, SSO[3] is OFF.
SRVO-409 DCS SSO Servo Disconnect	In DCS Safe I/O connect function, SSO[4] is OFF.
SRVO-410 DCS SSO NTED input	In DCS Safe I/O connect function, SSO[5] is OFF.
SRVO-419 DCS PROFIsafe comm. error	PROFINET Safety communication error occurs.

Smooth stop is different from **Power-Off stop** as follows:

- In Smooth stop, the robot is stopped along the program path. This function is effective for a system where the robot can interfere with other devices if it deviates from the program path.
- In Smooth stop, physical impact is less than Power-Off stop. This function is effective for systems where the physical impact to the mechanical unit or EOAT (End Of Arm Tool) should be minimized.
- The stopping distance and time of Smooth stop is longer than those of Power-Off stop, depending on the robot model and axis.

Smooth stop is different from **Controlled stop** as follows:

- The stopping distance and time of Smooth stop is normally shorter than those of Controlled stop, depending on the robot model and axis.

When this option is loaded, this function cannot be disabled.

The stop type of DCS Position and Speed Check functions is not affected by the loading of this option.

**WARNING**

The stopping distance and time of Smooth stop are longer than those of Power-Off stop. A risk assessment for the whole robot system which takes into consideration the increased stopping distance and stopping time, is necessary when this option is loaded.

9 STOP TYPE OF ROBOT (R-30iB Plus, R-30iB Mate Plus)

There are following three types of Stop Category.

Stop Category 0 following IEC 60204-1 (Power-off Stop)

Servo power is turned off, and the robot stops immediately. Servo power is turned off when the robot is moving, and the motion path of the deceleration is uncontrolled.

“**Stop Category 0**” performs following processing.

- An alarm is generated, and then the servo power turns off. Instantly the robot stops.
- Execution of the program is paused.

Frequent Category 0 Stop of the robot during operation can cause mechanical problems of the robot.

Avoid system designs that require routine or frequent Category 0 Stop conditions.

Stop Category 1 following IEC 60204-1 (Controlled Stop, Smooth Stop)

The robot is decelerated until it stops, and servo power is turned off.

“**Stop Category 1**” performs following processing.

- The alarm "**SRVO-199 Controlled stop**" or "**SRVO-289 Smooth Stop**" occurs along with a decelerated stop. The program execution is paused.
- An alarm is generated, and then the servo power turns off.

In Smooth stop, the robot decelerates until it stops with the deceleration time shorter than Controlled stop.

The stop type of Stop Category 1 is different according to the robot model or option configuration. Please refer to the operator's manual of a particular robot model.

Stop Category 2 following IEC 60204-1 (Hold)

The robot is decelerated until it stops, and servo power remains on.

“**Stop Category 2**” performs following processing.

- The robot operation is decelerated until it stops. Execution of the program is paused.

WARNING

- 1 The stopping distance and time of Stop Category 1 are longer than those of Stop Category 0. A risk assessment for the whole robot system which takes into consideration the increased stopping distance and stopping time is necessary when Stop Category 1 is used. Please refer to the operator's manual of a particular robot model for the data of stopping distance and time.
- 2 In multi arm system, the longest stopping distance and time of Stop Category 1 among each robot are adopted as those for the system. A risk assessment for the whole robot system which takes into consideration a possibility that the stopping distance and time increase, is necessary on the multi arm system.
- 3 In the system which has extended axis, the longer stopping distance and time of Stop Category 1 among robot and extended axis are adopted as those for the system. A risk assessment for the whole robot system which takes into consideration a possibility that the stopping distance and time increase, is necessary on the system which has extended axis. Please refer to the extended axis setup procedure of the controller operator's manual for considering the stopping distance and time of the extended axis.
- 4 When Stop Category 1 occurs during deceleration by Stop Category 2, the stop type of robot is changed to Stop Category 0.
- 5 In case of Stop Category 1, motor power shutdown is delayed for a maximum of 2 seconds. In this case, a risk assessment for the whole robot system is necessary, including the 2 seconds delay.

When the emergency stop button is pressed or the FENCE is open, the stop type of robot is Stop Category 0 or Stop Category 1. The configuration of stop type for each situation is called *stop pattern*. The stop pattern is different according to the option configuration.

There are the following 3 Stop patterns.

Stop pattern	Mode	Emergency stop button	External Emergency stop	FENCE open	SVOFF input	Deadman switch (*)
A	AUTO	Category 0	Category 0	Category 1	Category 1	-
	T1	Category 0	Category 0	-	Category 1	Category 0
	T2	Category 0	Category 0	-	Category 1	Category 0
C	AUTO	Category 1	Category 1	Category 1	Category 1	-
	T1	Category 0	Category 0	-	Category 1	Category 0
	T2	Category 0	Category 0	-	Category 1	Category 0
D	AUTO	Category 1	Category 1	Category 1	Category 1	-
	T1	Category 1	Category 1	-	Category 1	Category 1
	T2	Category 1	Category 1	-	Category 1	Category 1

Category 0: Stop Category 0

Category 1: Stop Category 1

-: Disable

(*) The stop pattern of NTED input is same as Deadman switch.

The following table indicates the Stop pattern according to the controller type or option configuration.

The case R651 is specified.

Option	R-30iB Plus/ R-30iB Mate Plus
Standard	C(**)
Old Stop Function (A05B-2670-J680)	A(**)
All Smooth Stop Function (A05B-2670-J651)	D(**)

The case R650 is specified.

Option	R-30iB Plus/ R-30iB Mate Plus
Standard	A(**)
Stop Category 1 by E-Stop (A05B-2670-J521)	C(**)
All Smooth Stop Function (A05B-2670-J651)	D(**)

(**)R-30iB Mate Plus does not have SVOFF input.

The stop pattern of the controller is displayed in "Stop pattern" line in software version screen. Please refer to "Software version" in operator's manual of controller for the detail of software version screen.

"Old Stop Function" option

When "Old Stop Function" (A05B-2670-J680) option is specified, the stop type of the following alarms becomes Stop Category 0 in AUTO mode.

Alarm	Condition
SRVO-001 Operator panel E-stop	Operator panel emergency stop is pressed.
SRVO-002 Teach pendant E-stop	Teach pendant emergency stop is pressed.
SRVO-007 External emergency stops	External emergency stop input (EES1-EES11, EES2-EES21) is open.
SRVO-408 DCS SSO Ext Emergency Stop	In DCS Safe I/O connect function, SSO[3] is OFF.
SRVO-409 DCS SSO Servo Disconnect	In DCS Safe I/O connect function, SSO[4] is OFF.

Stop Category 0 is different from **Stop Category 1** as follows:

- In Stop Category 0, servo power is turned off, and the robot stops immediately. Servo power is turned off when the robot is moving, and the motion path of the deceleration is uncontrolled.
- The stopping distance and time of Stop Category 0 is shorter than those of Stop Category 1, depending on the robot model and axis.

When this option is loaded, this function cannot be disabled.

The stop type of DCS Position and Speed Check functions is not affected by the loading of this option.

"All Smooth Stop Function" option

When "All Smooth Stop Function" (A05B-2670-J651) option is specified, the stop type of the following alarms becomes Stop Category 1 in all operation modes (AUTO, T1 and T2 mode).

Alarm	Condition
SRVO-001 Operator panel E-stop	Operator panel emergency stop is pressed.
SRVO-002 Teach pendant E-stop	Teach pendant emergency stop is pressed.
SRVO-003 Deadman switch released	Both deadman switches on Teach pendant are released.
SRVO-007 External emergency stops	External emergency stop input (EES1-EES11, EES2-EES21) is open.
SRVO-037 IMSTP input (Group: %d)	IMSTP input (*IMSTP signal for a peripheral device interface) is ON.
SRVO-232 NTED input	NTED input (NTED1-NTED11, NTED2-NTED21) is open.
SRVO-408 DCS SSO Ext Emergency Stop	In DCS Safe I/O connect function, SSO[3] is OFF.
SRVO-409 DCS SSO Servo Disconnect	In DCS Safe I/O connect function, SSO[4] is OFF.
SRVO-410 DCS SSO Ext Emergency Stop	In DCS Safe I/O connect function, SSO[5] is OFF.
SRVO-419 DCS PROFIsafe comm. error	PROFINET Safety communication error occurs.

Stop Category 1 is different from **Stop Category 0** as follows:

- In Stop Category 1, the robot is stopped along the program path. This function is effective for a system where the robot can interfere with other devices if it deviates from the program path.

- In Stop Category 1, physical impact is less than Stop Category 0. This function is effective for systems where the physical impact to the mechanical unit or EOAT (End of Arm Tool) should be minimized.
- The stopping distance and time of Stop Category 1 is longer than those of Stop Category 0, depending on the robot model and axis.

When this option is loaded, this function cannot be disabled.

The stop type of DCS Position and Speed Check functions is not affected by the loading of this option.



WARNING

The stopping distance and time of Stop Category 1 are longer than those of Stop Category 0. A risk assessment for the whole robot system which takes into consideration the increased stopping distance and stopping time, is necessary when this option is loaded.

"Stop Category 1 by E-Stop" option

When "Stop Category 1 by E-Stop" (A05B-2670-J521) option is specified, the stop type of the following alarms become Category 1 Stop but only in AUTO mode. In T1 or T2 mode, the stop type is Category 0 Stop which is the normal operation of the system.

Alarm	Condition
SRVO-001 Operator panel E-stop	Operator panel emergency stop is pressed.
SRVO-002 Teach pendant E-stop	Teach pendant emergency stop is pressed.
SRVO-007 External emergency stops	External emergency stop input (EES1-EES11, EES2-EES21) is open.
SRVO-408 DCS SSO Ext Emergency Stop	In DCS Safe I/O connect function, SSO[3] is OFF.
SRVO-409 DCS SSO Servo Disconnect	In DCS Safe I/O connect function, SSO[4] is OFF.

Stop Category 1 is different from **Stop Category 0** as follows:

- In Stop Category 1, the robot is stopped along the program path. This function is effective for a system where the robot can interfere with other devices if it deviates from the program path.
- In Stop Category 1, physical impact is less than Stop Category 0. This function is effective for systems where the physical impact to the mechanical unit or EOAT (End of Arm Tool) should be minimized.
- The stopping distance and time of Stop Category 1 is longer than those of Stop Category 0, depending on the robot model and axis.

When this option is loaded, this function cannot be disabled.

The stop type of DCS Position and Speed Check functions is not affected by the loading of this option.



WARNING

The stopping distance and time of Stop Category 1 are longer than those of Stop Category 0. A risk assessment for the whole robot system which takes into consideration the increased stopping distance and stopping time, is necessary when this option is loaded.

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1 INTRODUCTION

This manual primarily describes the operation on the robot teach pendant for the Integrated Welding Controller.

For installation, timer functions, and timer error troubleshooting, of the timer control board, refer to the Robot Integrated Weld Control for Fanuc R-30iA/R-30iB, issued by DENGENSHA MFG. CO., LTD.

NOTE

- In this manual, the timer control board for the above integrated welding controller will be referred to as the timer board or the integrated timer board.
- This feature is an option.

2 SETUP

2.1 SETTING UP WELD INTERFACE

A weld interface (Digital or Integral) can be confirmed and/or changed on the Spot Config screen. The default value of a weld interface is "Integral" if your robot has this software option.

When changing a weld interface, or when increasing the number of equipment, there is a need to change items on this screen

Operation

MENUS → SETUP → F1 [TYPE] → Spot Config

The actual screen is different from each equipment type (air gun or servo gun). The following is servo gun example.

SPOT CONFIG	
	6/34
1 F Number:	F00000
2 Load Spottool Macros:	DISABLED
3 Number of equipments:	1
4 Number of weld controllers:	1
5 Current equipment, gun:	1, 1
6 Weld interface:	INTEGRAL
7 Number of guns:	SINGLE
:	:
13 Weld cntler num for gun:	1
Current weld controller:	1
:	:
:	:

In the case of one integrated timer board

Item name	Setting value
Number of equipments	1 (default)
Number of weld controllers	1 (default)
Current equipment, gun	1,1 (*1)
Weld interface	INTEGRAL (*2)
Number of guns	1 (default)
Weld cntler num for gun	1 (default)

*1) When you want to set items for equipment 1 and gun 1, set this item 1,1.

*2) When your robot software does not have integrated weld controller option, you can not change this to INTEGRAL. If your robot software has integrated weld controller but you want to change to the un-integrated weld controller, change this item to DIGITAL.

In the case of two integrated timer boards

Item name	Setting value	
Number of equipments	2	
Number of weld controllers	2	
Current equipment, gun	1,1	2,1
Weld interface	INTEGRAL	INTEGRAL
Number of guns	1	1
Weld cntler num for gun	1	2

NOTE

- If one timer board is installed, set both T0 and T1 of the DIP switch [SEL] on the timer board to 0.
- If two timer boards are installed, set T0 and T1 to 0 on the first timer board, and set T0 to 1 and T1 to 0 on the second timer board.
- Refer to the Robot Integrated Weld Control for Fanuc R-30iA/ R-30iB, issued by DENGSHA MFG. CO., LTD. for more details.

2.2 SETTING UP SIGNALS

You must assign the following signals according to the actual wiring. About the necessity of setup and signal number, it is dependent on the contactor specification and user PLC specification.

2.2.1 Weld Interface Signals

Operation

MENUS → I/O → F1 [TYPE] → Weld interface

Output signals (OUT)

I/O Weld Out						
WC: 1			1/4			
NAME	OUT	PT	SIM	STATUS		
1 Iso contactor:	DO[	0]	U	***		
2 Shunt trip:	DO[	0]	U	***		
3 Alarm code:	GO[	0]	U	***		
4 Tip stick timing:	DO[	0]	U	***		
	[TYPE]		IN/OUT	ON	OFF	>

Press F3 to switch between IN and OUT

Input signals (IN)

I/O Weld IN						
WC: 1			1/3			
NAME	IN	PT	SIM	STATUS		
1 Iso contactor:	DI[	0]	U	***		
2 Leak input:	DI[	0]	U	***		
3 Tip stick detect:	DI[	0]	U	***		
	[TYPE]		IN/OUT	ON	OFF	>

NOTE

After changing any settings, turn the power off, then on again.

Each signal is described below.

Output signals (OUT)

Item	Description
1.Iso contactor (IC close signal)	Sets the number of the output signal to the IC close signal on the contactor unit. Note) Install wiring so that the contactor is closed when this signal is on.

Item	Description
2.Shunt trip (CB shunt trip)	Sets the number of the output signal to the CB shunt trip signal on the contactor unit. This signal is depend on the actual contactor specification. This signal is output by a shunt trip request from the timer board. The circuit breaker on the contactor unit is tripped by the pulse output. A directive for tripping the circuit breaker is transferred from the timer board to the robot, then to the contactor unit.
3.Alarm code	Signal for outputting a timer board alarm I/O code to peripherals. Set a GO (group output) 4 bits. The signals to be used additionally require group output (GO) assignment.
4.Tip stick timing	In servo gun system, this signal can be used to inform the PLC when the tip stick check was performed. Each time the gun opens from the weld spot, the tip stick timing output is turned on when the gun reaches the opening specified by the Tip Stick Detect Distance. This output turns on regardless of whether the tips are stuck or not.

Input signals (IN)

Item	Description
1.Iso contactor (IC auxiliary contact output signal)	Signal for detecting the contactor open/close status. Set this signal so that it is on when the contactor is closed or off when the contactor is open.
2.Leak Input	Signal for detecting a leakage of current. When this signal is turned on, 'SPOT-384 IWC(DG) leak input' occurs in the robot.
3.Tip stick detect	In servo gun system, this signal is used for tip stick detection by weld controller. Robot reads this input at gun open after welding, and if this signal is on at tip stick detect distance, robot stops and post the tip stick detection alarm.

NOTE

Whether signal setup is required or not depends on the specifications of the contactor. When a signal need not be wired, set its signal number to 0.
Also when direct control from the robot controller is not required, set signal numbers to 0.

When the contactor signals are to be used, check the following after completing wiring and setting.

Check 1 Turn on contactor DO.
The contactor(MC) of the contactor unit closes.
Contactor DI is turned on.

Check 2 Turn off contactor DO.
The contactor(MC) opens.
Contactor DI is turned off.

NOTE

The conditions for turning the contactor output signal on and off can be changed by setting the contactor control type on the spot config screen. For details, refer to the R-30iA Controller Spot Tool+ operator's manual (B-83124EN-1) or R-30iB Controller Spot Welding Function operator's manual (B-83284EN-4). When DENGENSHA's integrated timer is specified, the contactor output signal is synchronized with the on/off setting of the Weld/NOWELD input signal of cell I/O.

2.2.2 IWC Communication Status Signal

Operation

MENUS → I/O → F1 [TYPE] → Cell Intface

Press F3, IN/OUT to open the Cell outputs screen.

I/O Cell Outputs				
				1/21
NAME	OUT	PT	SIM	STATUS
1 Input Simulated	DO[	0]	U	***
2 Output Simulated	DO[	0]	U	***
3 OVERRIDE = 100	DO[	0]	U	***
4 In cycle	DO[	0]	U	***
5 Prog Aborted	DO[	0]	U	***
6 Process fault	DO[	0]	U	***
7 Process alert	DO[	0]	U	***
8 Process complete	DO[	0]	U	***
9 Weld enabled	DO[	0]	U	***
10 Stroke enabled	DO[	0]	U	***
11 Proc1 tip rep re	DO[	0]	U	***
12 Proc2 tip rep re	DO[	0]	U	***
13 Apprch tip repla	DO[	0]	U	***
14 Proc1 tip mnt re	DO[	0]	U	***
15 Proc2 tip mnt re	DO[	0]	U	***
16 One Spot Welded	DO[	0]	U	***
17 Tryout Status	DO[	0]	U	***
18 Heartbeat	DO[	0]	U	***
19 IWC COMM Stat	DO[	0]	U	***
20 Robot motion G1	DO[	0]	U	***
21 Robot motion G2	DO[	0]	U	***
	[TYPE]		IN/OUT	SIM UNSIM >

NOTE

The actual item number is different from each system configuration.
If you can not see the IWC COMM stat, check the weld interface in the SETUP Spot Config screen, and after that enter the function key REDO on this screen.

Item	Description
IWC COMM stat	When this signal is set to ON, communication with the timer board has been established. When the signal is OFF, communication with the timer board is not established. If you install two timer boards, this signal is ON if and only if the system is in communication with both boards.

2.2.3 Output Signal to External Equipment

A part of the communication data between timer board and robot can be output to external equipment by using digital I/O (DOUT or GOUT). The data that can be output are Weld Complete, Weld Schedule, Weld Sequence in Progress, and Weld in Progress. Follow the following procedures when you want to output these data to external equipment (PLC etc.).

Operation

MENUS → SYSTEM → F1 [TYPE] → Variables

Change the following system variable to TRUE.

\$IWCPCFG.\$EXTEND_ENB2[#] = TRUE ENABLED
\$IWCPCFG.\$EXTEND_ENB2[#] = FALSE DISABLED (Default)
: Weld controller number

NOTE

Re-restore power supply after changing the above-mentioned system variable.

Set each signal by the following system variables.

Name	System variable	Description
Weld Complete	\$IWPCCFG.\$EXTEND2[#].\$OT_IDX[1] # : Weld controller number	Specify port number of DOUT for this system variable. When weld sequence is completed, pulse of 0.5msec is output. When weld sequence ends on the way, this is not output. Change \$IWPCCFG.\$EXTEND2[#].\$TIME[1] when you want to change width of pulse.
Weld Schedule	\$IWPCCFG.\$EXTEND2[#].\$OT_IDX[2] # : Weld controller number	Specify port number of GOUT for this system variable. Weld schedule is output while executing the weld sequence.
Weld Sequence in Progress	\$IWPCCFG.\$EXTEND2[#].\$OT_IDX[3] # : Weld controller number	Specify port number of DOUT for this system variable. This signal is turned on while executing the weld sequence.
Weld in Progress	\$IWPCCFG.\$EXTEND2[#].\$OT_IDX[4] # : Weld controller number	Specify port number of DOUT for this system variable. This signal is turned on when energizing while executing the weld sequence.

NOTE

Re-restore power supply after changing the above-mentioned system variables.

NOTE

- 1 Weld Complete, Weld Schedule, and Weld Sequence in Progress are output irrespective of weld mode (Weld/No weld). Observe Weld in Progress also when you observe energizing.
- 2 Weld in Progress is turned on with WELD in the welding sequence, and turned off with HOLD. This is not turned off in COOL etc.
- 3 Even if next spot instruction is executed while the pulse of Weld Complete has been turned on, the pulse of Weld Complete is not turned off.

NOTE

This function has been supported since software version 7DC1P06.

2.3 SETTING UP THE INTEGRATED TIMER PARAMETERS

Settings for the integrated timer board can be made and confirmed from the robot teach pendant.

NOTE

For the items on this setting screen, be sure to refer to the manual issued by DENGSHA MFG. CO., LTD. to make settings correctly.

Operation

MENUS → 0 NEXT → DEP(DG) → F1 [TYPE] → Setup

DEP Setup	
	1/5
WC:	1
1 Common	<*DETAIL*>
2 Schedule	<*DETAIL*>
3 Gun	<*DETAIL*>
4 Function	<*DETAIL*>
5 Hardware	<*DETAIL*>

	[TYPE]					>
--	--------	--	--	--	--	---

Move the cursor on <*DETAIL*> and press the ENTER key to display the detail setup screen for the corresponding item.

The screen display is returned from each detail setup screen to the above screen by pressing the PREV key.

These screens correspond to the following welding parameters.

Common ... Sequence C (Common sequence)
 Schedule ... Schedule 1-255 (Weld Schedules)
 Gun ... Sequence G1-4
 Function ... Sequence S (Special Function)
 Hardware ... Sequence H (Hardware, Gain)

When two timer boards are used, switching to it by using F3 WC_NUM on each screen. If your system has only a timer board, F3 WC_NUM is not displayed.

NOTE

Do not change any value on each screen during welding execution.

2.3.1 COMMON (Common)

Operation

MENUS → 0 NEXT → DEP(DG) → F1[TYPE] → Setup → 1 common

DEP COMMON	
WC: 1	1/8
1 WELD TIME COMP (CYC)	0
2 WELD TIME SHORT (CYC)	0
3 WELD TIME LIMIT (CYC)	1000
4 REF LINE VOLT (V)	400
5 LINE VOLT LOW LIMIT (%)	20
6 LINE VOLT HIGH LIMIT (%)	20
7 W-CUR FAULT LIMIT (%)	30
8 OVER CURRENT LIMIT (%)	120

	[TYPE]		WC_NUM			>
--	--------	--	--------	--	--	---

To change a setting, move the cursor on the target item, enter the setting value, then press the ENTER key.

For the meanings and input ranges of the displayed items, refer to the manual issued by DENGENSHA MFG. CO., LTD.

Function key

F3 WC_NUM	Changes the target weld controller number for display. This function key is displayed when more than one timer board is installed.
-----------	--

2.3.2 WELD SCHEDULE

Operation

MENUS → 0 NEXT → DEP(DG) → F1 [TYPE] → Setup → 2 Schedule

DEP SCHEDULE					
WC: 1 Schedule: 1		1/34			
1	SQUEEZE (CYC)	50			
2	UP-SLOPE (CYC)	3			
3	WELD 1 (CYC)	0			
4	COOL1 (CYC)	0			
5	WELD 2 (CYC)	0			
6	COOL2 (CYC)	0			
7	WELD 3 (CYC)	0			
8	DOWN-SLOPE (CYC)	0			
9	HOLD (CYC)	20			
10	MULTI IMPULSE	1			
11	CURRENT 1 (A)	2000			
12	CURRENT 2 (A)	2000			
13	CURRENT 3 (A)	2000			
14	HEAT 1 (%)	20			
15	HEAT 2 (%)	20			
16	HEAT 3 (%)	20			
17	CURRENT HIGH LIMIT (%)	20			
18	CURRENT LOW LIMIT (%)	20			
19	HEAT HIGH LIMIT (%)	100			
20	HEAT LOW LIMIT (%)	10			
21	TRANS TURNS	200.0			
22	CTRL(VC=0,CC,CP,TWCP)	1			
23	MAX CURRENT (A)	60000			
24	STEP 1 CURRENT (%)	100			
25	STEP 2 CURRENT (%)	100			
26	STEP 3 CURRENT (%)	100			
27	STEP 4 CURRENT (%)	100			
28	STEP 5 CURRENT (%)	100			
29	STEP 6 CURRENT (%)	100			
30	STEP 7 CURRENT (%)	100			
31	STEP 8 CURRENT (%)	100			
32	STEP 9 CURRENT (%)	100			
33	GUN No (1-4)	1			
34	MONITOR IMPULSE (1-3)	1			
	[TYPE]	SCHE_NUM	WC_NUM	PREV	NEXT >
	SAVE	LOAD	PRINT	COPY	

WC:
Target weld controller
(= timer board) number.

Schedule:
Target welding schedule number.

The screen is displayed based on the information from the timer board. To change displayed data, move the cursor on the target item, enter the setting value, then press the ENTER key.

For the meanings and input ranges of the displayed items, refer to the manual issued by DENGENSHA MFG. CO., LTD.

Function keys

Page 1

F2 SCHE_NUM	Changes the weld schedule number.
F3 WC_NUM	Changes the weld controller number. The weld controller number is the number of a timer board. This function key is displayed when more than one timer board is installed.
F4 PREV	Displays the previous weld schedule.
F5 NEXT	Displays the next weld schedule.

Page 2

F1 SAVE	Saves a schedule in text form. The file name is "weld-controller-number"sche"weld-schedule-number".dt. For example, in the case of weld controller 1 and weld schedule 10, the file name is 1sch10.dt. For details, see section 5.2. For how to save and load data using the File screen, see Section 5.1.
---------	--

F2 LOAD	Loads the weld schedule in a file named as described above. For details, see section 5.2.
F3 PRINT	Outputs the displayed schedule in text form just as displayed on the screen. The file name is "weld controller-number"sche "weld schedule-number".ls. For example, in the case of weld controller 1 and weld schedule 5, the file name is 1sch5.ls. You cannot upload or download data via FTP.
F4 COPY	You can copy the currently displayed weld schedule to another weld schedule number.

2.3.3 COUNT/STEP (Gun)

Operation

MENUS → 0 NEXT → DEP(DG) → F1 [TYPE] → Setup → 3 Gun

DEP COUNT						
WC: 1 GUN: 1			1/12			
1	TIP DRESS COUNT					0
2	TIP CHANGE COUNT					0
3	STEPPER STEP 0 COUNT					0
4	STEPPER STEP 1 COUNT					0
5	STEPPER STEP 2 COUNT					0
6	STEPPER STEP 3 COUNT					0
7	STEPPER STEP 4 COUNT					0
8	STEPPER STEP 5 COUNT					0
9	STEPPER STEP 6 COUNT					0
10	STEPPER STEP 7 COUNT					0
11	STEPPER STEP 8 COUNT					0
12	STEPPER STEP 9 COUNT					0
	[TYPE]	GUN_NUM	WC_NUM	PREV	NEXT	>

The screen is displayed based on the information from the timer board. To change displayed data, move the cursor on the target item, enter a value, then press the ENTER key.

For the meanings and input ranges of the displayed items, refer to the manual issued by DENGENSHA MFG. CO., LTD.

Function keys

F2 GUN_NUM	Changes the number of the gun to be displayed.
F3 WC_NUM	Changes the weld controller number. The weld controller number is the number of a timer board. This function key is displayed when more than one timer board is installed.
F4 PREV	Changes to the previous gun.
F5 NEXT	Changes to the next gun.

2.3.4 FUNCTION (Special)

Operation

MENUS → 0 NEXT → DEP(DG) → F1 [TYPE] → Setup → 4 Function

DEP FUNCTION					
WC: 1			1/7		
1	REWELD		OFF		
2	FAULT RESET BY PILOT		OFF		
3	ALARM LEVEL		ALARM		
4	STEPPER		OFF		
5	TOROID SENSITIVITY		STD		
6	PWM FREQUENCY		600		
7	INV CAPACITY		400		
	[TYPE]		WC_NUM	[CHOICE]	>

The screen is displayed based on the information from the timer board. To change displayed data, move the cursor on the target item, and press F4 [CHOICE] key.

For the meanings and input ranges of the displayed items, refer to the manual issued by DENGENSHA MFG. CO., LTD.

Function key

F3 WC_NUM	Changes the weld controller number. The weld controller number is the number of a timer board. This function key is displayed when more than one timer board is installed.
-----------	--

2.3.5 HARD (Hardware)

Operation

MENUS → 0 NEXT → DEP(DG) → F1 [TYPE] → Setup → 5 Hardware

DEP HARD					
WC: 1			1/6		
1	PRIM-CT GAIN		10000		
2	2ND TOROID GAIN		10000		
3	LINE VOLT GAIN		10000		
4	TIP RESISTANCE GAIN		10000		
5	FIXED ARM GAIN		10000		
6	MOVABLE ARM GAIN		10000		
	[TYPE]		WC_NUM	UPDATE	>

The screen is displayed based on the information from the timer board. To change displayed data, move the cursor on the target item, enter the setting value, then press the ENTER key.

To make the new value effective, press F4, UPDATE.

For the meanings and input ranges of the displayed items, refer to the manual issued by DENGENSHA MFG. CO., LTD.

Function keys

F3 WC_NUM	Changes the weld controller number. The weld controller number is the number of a timer board. This function key is displayed when more than one timer board is installed.
F4 UPDATE	Writes a new value into the timer board.

NOTE

This hardware data is the timer board circuit compensation data. And it is factory-set by DENGENSHA. You need not change any setting normally. The data is retained not on the robot memory, but only is in the timer board memory.

3 PROGRAM INSTRUCTIONS

3.1 SPOT INSTRUCTION

The SPOT instruction directs the timer board to start welding according to the specified weld schedule. After completion of welding, the next line in the program is executed:

SPOT [...S=1...] (“...” is different with equipment type)

S: Weld schedule number

3.2 RESET STEPPER INSTRUCTION

The RESET STEPPER instruction resets stepper counters, tip dress counter, and tip replace counter to 0.

RESET STEPPER [WC=1, SN=1, SV=0]

WC : Timer board number (1 or 2)

SN : G1-4 series (1 to 4)

SV : Option

0: Clears the "stepper counters" and "stepper step counts" for the gun specified by SN to 0.

1: Clears the "tip dress counter" for the gun specified by SN to 0.

2: Clears the "tip replace counter" for the gun specified by SN to 0.

3.3 RESET WELDER INSTRUCTION

The RESET WELDER instruction is used to reset a welding error on the welding timer board from the robot controller.

RESET WELDER [WC=weld controller-number]

NOTE

This instruction resets error only on the timer board. If you enable “Reset WC with Robot” on the spot config screen, the timer board alarms are reset by reset operations (teach pendant reset, UI[reset], or SI[reset]).

4 ALARM/RESET

You can check timer board alarms on the following two screens:

- Alarm/Hist screen
- DEP/Alarm

- Alarm/Hist screen

You can check timer board alarms together with other alarms. See section 4.1 for details

- DEP/Alarm screen, Alarm/Hist screen

You can check alarm messages posted only from the timer board and an alarm history for each alarm level. See section 4.2 for details.

For resetting alarms of the timer board, see section 4.4.

4.1 ALARM/HIST SCREEN

Operation

MENUS → Alarm → F3 HIST

(Display example)

Alarm : Hist					
DETAIL Alarm					
spot-010 Major alarm detected					
STOP.G 12-MAY-06 23:58					
ALARM : HIST					
1 spot-010 Major alarm detected					
2 spot-064 CURRENT LOW					
3 spot-064 TOTAL W-TIME LONG					
4 RESET					
	[TYPE]	[VIEW]	ACTIVE	CLEAR	DETAIL

More than one alarm message may be issued just by one spot welding.
In the above example, current low and total w-time long are posted.

- **SPOT-010 Major alarm detected <robot stopped>**
If any faults (fault or error) are included in an alarm message posted during welding, this alarm is issued, and the robot is stopped.
- **SPOT-011 Minor alarm detected <warn, not stopped>**
If no fault (fault or error) is included in an alarm message posted during welding and the alarm message is just a warning (alarm), this warning alarm is issued.
- Posted alarm messages can be checked with the alarm history.
SPOT-063 "alarm-message" stopped
SPOT-064 "alarm-message" stopped or warning
- On the DEP(DG)/alarm screen (see Section 4.2), alarm messages issued only from the timer can be checked.

NOTE

For cause and action against alarm messages from the timer board, refer to the manual issued by DENGENSHA MFG. CO., LTD.

When shell polling is enabled and program start from TP, if alarm is received from timer board, spot-010 or spot-011 is posted twice.

Relation with alarm level of timer board and robot stop

Timer board alarm level	Robot
ERROR level	Pause (STOP.G)
FAULT level	
ALARM level	None pause (WARN)

Alarm message, I/O code, alarm level

Alarm message	I/O code (*1)	Alarm level (*2)
CURRENT LOW	1	Alarm
CURRENT HIGH	2	Alarm
HEAT %LOW	3	Alarm
HEAT %HIGH	4	Alarm
LINE VOLT LOW	5	Alarm
LINE VOLT HIGH	6	Alarm
STEP END	7	Alarm
TIP DRESS END		Alarm
TIP CHANGE END		Alarm
LINE CLOCK	8	Fault
NO CURRENT	9	Fault
CURRENT FAULT	10	Fault
TIME FAULT		Fault
I/O FAULT		Fault
TRANS OVER TEMP	11	Fault
SCR OVER TEMP		Fault
SCR SHORT	12	Fault
SCR HALF CYCLE		Fault
MEMORY ERROR	15	Error
W-CUR PROGRAM LOW		Error
W-CUR PROGRAM HIGH		Error
TOTAL W-TIME LONG		Error
PARAMETER OUT LIMIT		Error

NOTE

The information on an alarm message, an I/O code, and an alarm level is associated with the timer board to be used. Refer to the manual issued by DENGENSHA MFG. CO., LTD.

(*1) The I/O code can be output to the group signal output. For how to set the signal, see Section 2.2 "SETTING UP SIGNALS.". The above I/O code indicates the value output for the corresponding alarm.

(*2) Some alarm levels can be changed into a FAULT level by the DEP / setup / special function screen.

4.2 INTEGRATED TIMER ALARM SCREEN

Operation

MENUS → 0 NEXT → DEP(DG) → F1 [TYPE] → Alarm

Fault						
WC: 1						
Message				LEVEL		
1 CURRENT LOW				Alarm		
2 TOTAL W-TIME LONG				Error		
	[TYPE]	RESET	WC_NUM	HIST		>

This screen displays the latest alarms on timer board.

This screen displays max four alarms that have not been reset.

Function keys

F2 RESET	Resets the timer board to release timer board alarms. This function key does not reset robot alarms.
F3 WC_NUM	Changes the weld controller number. The weld controller number is the number of a timer board. This function key is displayed when more than one timer board is installed.
F4 HIST	Shows past alarms issued from the timer board and the number of times such alarms were issued. (See Section 4.3)

After the timer board alarms are reset, the following screen is displayed:

Fault						
WC: 1						
Message				LEVEL		
NO ACTIVE ALARM						
	[TYPE]	RESET	WC_NUM	HIST		>

The PREV key returns the screen display to the above screen when you enter the history screen by using F4, HIST.

4.3 INTEGRATED TIMER/ ALARM HIST SCREEN

Operation

MENUS → 0 NEXT → DEP(DG) → F1 [TYPE] → Alarm → F4 HIST

Fault Frequency						
ALL/ALL				1/4		
Fault				# of Occur		
1 CURRENT LOW				2		
2 LINE VOLT HIGH				3		
3 TIME FAULT				1		
4 W-CUR PROGRAM LOW				1		
	[TYPE]		PART_ID	SAVE		>

This screen displays alarm history which is issued during previous welding. From this screen, you can display the number of alarms for each program or position and details of each alarm.

Press F3, PART_ID. The following screen appears:

Part Fault Freq.						
ALL/ALL		1/2				
	PART ID/	Program name	# of Faults			
1	AA_01		6			
2	AA_02		3			
	[TYPE]		WELD_ID	SAVE		>

You can check the number of alarms for each program.

NOTE

If you select STYLE program select mode, this screen displays the number of alarms for each style id.

Press F3, WELD_ID. The following screen appears:

Weld Fault Freq.						
AA_02 /ALL		1/2				
	Weld ID	# of Faults				
1	Pos.1	2				
2	Pos.2 (Body side2)	1				
	[TYPE]		FLTS	SAVE		>

You can check the number of alarms for each position with the specified program.

NOTE

This screen does not display the number of alarms for a position numbered 1000 or greater.

Press F3, FLTS. The following screen appears:

Weld Faults						
AA_02/1		1/2				
	Weld ID	# of Occur				
1	CURRENT LOW	1				
2	LINE VOLT HIGH	1				
	[TYPE]			SAVE		>

You can check the number of alarms and type of each alarm for the specified program and position.

Press F4, SAVE to save the welding results and welding alarm history. (See section 7.3)

4.4 ALARM RESET FOR INTEGRATED TIMER

If you enable “Reset WC with Robot” on the spot config screen, the timer board alarms are reset by reset operations (teach pendant reset, UI[reset], or SI[reset]).

Reset WC with Robot on SETUP/Spot Config

ENABLED : The timer board alarms are reset by reset operations of robot (TP, UI, or SI). (default)

DISABLED: The timer board alarms are not reset by reset operations of robot (TP, UI, or SI).

When the above setting is disabled, you can reset the timer board using the methods listed below:

1. Function key(F2) RESET of DEP(DG)/Alarm (See section 4.2)
2. Program instruction (RESET WELDER) (See section 3.3)

NOTE

When Reset WC with Robot is disabled, both reset operation of a timer board and reset operation of a robot may be needed.

5 SAVING AND LOADING

5.1 SAVING SCHEDULE AND SETUP DATA

The data listed below is saved into the following SV files. The files are saved by [all of above] or [System files] on the File screen. To load the files, use the controlled start screen.

Operation

MENUS → File → F4 [BACKUP] → [All of above] or [System files]

IWDGSCHE.SV

Weld Schedule (all weld schedules)

Number of weld schedule (\$IWCDGSNUM)

IWDGCONF.SV

COMMON (Common)

COUNT/STEP (Gun)

FUNCTION (Special)

Current values for the tip dress counter, stepper counter (You cannot set the screen.)

SYSSPOT.SV

Spot-related settings other than the above

NOTE

The settings on the HARD (Hardware) cannot be saved.

5.2 SAVE/LOAD WELD SCHEDULE WITH TEXT FORM

This section explains how to save/load a schedule to/from the file device with text form.

Operation

MENUS → 0 NEXT → DEP(DG) → F1 [TYPE] → Setup → 2 Schedule

DEP SCHEDULE						
WC: 1 GUN: 1				1/34		
1	SQUEEZE (CYC)			50		
2	UP-SLOPE (CYC)			3		
3	WELD 1 (CYC)			0		
:				:		
:				:		
32	STEP 9 CURRENT (%)			100		
33	GUN No (1-4)			1		
34	MONITOR IMPULSE (1-3)			1		
	[TYPE]	SCHE_NUM	WC_NUM	PREV	NEXT	>
	SAVE	LOAD	PRINT	COPY		

Press -> and F1 SAVE to save weld schedules with text form.

Press -> and F2 LOAD to load the text form's weld schedules.

DEP SCHEDULE	
WC: 1 GUN: 1	1/34
1 SQUEEZE (CYC)	50
2 UP-SLOPE (CYC)	3
3 WELD 1 (CYC)	0
4 COOL1 (CYC)	0
5 WELD 2 (CYC)	0
6 COOL2 (CYC)	0
7 WELD 3 (CYC)	0
8 DOWN-SLOPE (CYC)	0
9 HOLD (CYC)	20
10 MULTI IMPULSE	1
Select range (prev to exit)	
This sch	ALL SPECIFY
	>

Press one of F1 to F3.

F1 This Sch. : Saves(or loads) a currently selected weld schedule.

F2 ALL : Saves(or loads) all schedules. Each schedule is saved in a different file name.

F3 SPECIFY : Specifies the weld schedule range.

After pressing F3, specify the range as follows.

Input card number (0 is all):

You can specify the weld timer board number for target data.

When you use one weld timer board, this message does not appear.

Enter 0 to save(or load) schedules for all weld timer boards.

Enter 1 to save(or load) schedules for timer board 1.

Enter 2 to save(or load) schedules for timer board 2.

from (0 is all):

You can specify the target range of the weld schedule for the specified weld controller.

Specify the starting number for the save/load schedules.

Enter 0 to save/load all weld schedules.

to(0 for all after "from"):

Specify the number of the last weld schedule number you want to save or load.

Enter 0 to save (or load) all the subsequent weld schedules.

Enter a value greater than or equal to the starting number.

Examples (Total timer boards: 2, Total weld schedules: 255)

Specified range			Save/load range
Card	From	To	
1	2	5	Weld schedules 2 to 5 for timer board 1
2	8	0	Weld schedules 8 to 255 for timer board 2
1	7	7	Weld schedule 7 for weld controller 1
1	0	Not required	Weld schedules 1 to 255 for timer board 1
0	0	Not required	All weld schedules for timer board 1 and 2
0	8	0	Weld schedules 8 to 255 for timer board 1 and 2
0	2	5	Weld schedules 2 to 5 for timer board 1 and 2

When ALL or SPECIFY is selected, multiple files are saved or loaded. If a load or save error occurs during operation, processing stops at that point. The remaining files are not saved or loaded.

Example of a saved single weld schedule

This file was created on 22-OCT-05 17:54 from SpotTool+ V8.10xx

Robot F-Number:F00000

FILE:1sch32.dt

Robot Hostname:ROBOT

50,3,0,0,0,0,0,0,20,1,2000,2000,2000,

6 MONITORING WELDING RESULTS

You can check the welding results with the DEP(DG) Monitor screen.

6.1 MONITOR SCREEN

Operation

MENUS → 0 NEXT → DEP(DG) → F1 [TYPE] → Monitor

DEP MONITOR	
WC: 1	1/21
1 PART ID	AA_01
2 WELD ID	0
3 SCHEDULE No.	1
4 WELD CURRENT (A)	2000
5 WELD TIME (CYC)	0
6 HEAT (%)	50
7 POWER FACTOR (%)	60
8 LINE VOLT (V)	400
9 GUN No	1
10 TIP DRESS COUNT	0
11 TIP CHANGE COUNT	0
12 STEP No	0
13 WELDS COUNT	0
14 TIP RESISTANCE (m-ohm)	0
15 EXPANSION	0
16 FIXED ARM STRAIN	0
17 MOVABLE ARM STRAIN	0
18 Fault 1	0
19 Fault 2	0
20 Fault 3	0
21 Fault 4	0

The results of welding at the latest spot are displayed.

The displayed items differ depending on the option configuration of the timer board.

Press F4 LOG to display the DEP(DG) Data Log (previous data) screen.

NOTE

This screen does not display WELD ID for a position numbered 1000 or greater.

NOTE

For details of each item, refer to the manual issued by DENGENSHA MFG. CO., LTD.

6.2 DATA LOG

Operation

MENUS → 0 NEXT → DEP(DG) → F1 [TYPE] → Monitor → F4 [Log]

DEP Data Log				
				1/3
WC:			1	
1	PART ID(Program):	(ALL	)	
2	WELD ID:	0		
3	SCHEDULE:	0		
PARAM	Avg	Std	Var	
V :	400.00	0.00	0.00	
I :	2000.00	0.00	0.00	
PF :	60.00	0.00	0.00	
HEAT :	50.00	0.00	0.00	
[TYPE]		WC_NUM	[CHOICE]	ALL_PRG >

The results of previous welding at the latest 100 spots are recorded.

For voltage (V), current (I), power factor (PF), and heat ratio (HEAT), the average (Avg), standard deviation (std), and variance (var) are displayed.

Statistical data is displayed for the specified program and position.

(average, standard deviation, and variance)

When you specify WELD ID and SCHEDULE, the relevant data recorded for the 100 spots is used for statistical data processing and the resultant data is displayed. When you specify 0 for an item, the item is ignored

Program is selected from the sub window opened by pressing F4, [CHOICE]. When you press F5, ALL_PRG, the statistical data for all programs is displayed.

NOTE

This screen does not display a result for a position numbered 1000 or greater.

When you select STYLE program select mode, the following screen appears.

DEP Data Log				
				1/3
WC:			1	
1	PART ID(Program):		0	(* ALL *)
2	WELD ID:		0	
3	SCHEDULE:		0	
PARAM	Avg	Std	Var	
V :	400.00	0.00	0.00	
I :	2000.00	0.00	0.00	
PF :	60.00	0.00	0.00	
HEAT :	50.00	0.00	0.00	
[TYPE]	PART_ID	WC_NUM	MONITOR	>

Statistical data is displayed for a PART_ID and position number.

When you specify PART ID, WELD ID and SCHEDULE, the relevant data recorded for the 100 spots is used for statistical data processing and the resultant data is displayed. When you specify 0 for an item, the item is ignored. When you specify 0 for all items, statistical data for the 100 spots is processed and displayed.

You can also specify a value for PART ID as follows.

Press F2, PART_ID. The following screen appears.

DEP PART ID						
STYLE: * ALL *						1/32
ID	ProgName					
1	JOB123					
2	2_DOOR					
3	J67					
4	*****					
5	*****					
:	:					
:	:					
	[TYPE]	select				>

Move the cursor on the program for which you want to display data and press F2, SELECT.
The corresponding number is input for PART ID on the DEP Data Log screen.

7 OTHER

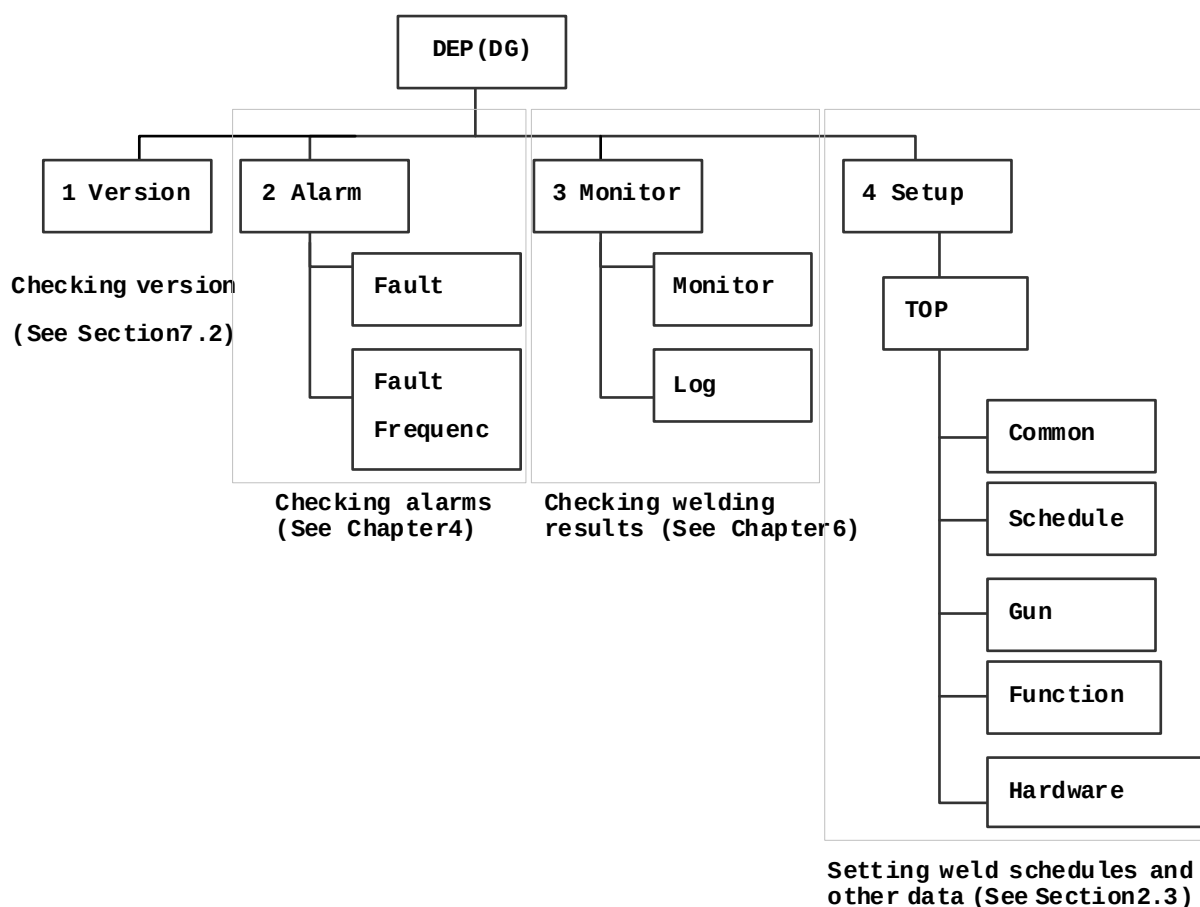
7.1 THE SCREEN CONFIGURATION OF DEP(DG) SCREEN

The settings related to the integrated timer are organized under

Operation

MENUS → 0 NEXT → DEP(DG).

Screen map



7.2 TIMER VERSION

You can confirm the version of timer board to following operation.

Operation

MENUS → 0 NEXT → DEP(DG) → F1 [TYPE] → Version

(Example)

Data Entry Panel						
Integrated Dengensha Welder						
timer software						
WC: 1 RWC2000 1003000200000003						
	[TYPE]					>

The following screens may be displayed immediately after a power cycle. Once change to another screen and display this screen again.

R-30iA

Data Entry Panel						
Integrated Dengensha Welder						
IWC(DG) is not ready						
	[TYPE]					>

R-30iB

Data Entry Panel						
Integrated Dengensha Welder						
Waiting for response from timer. SPOT alarm might occurred when there is no response even if waiting for a while. Confirm SPOT alarm on alarm screen.						
	[TYPE]					>

7.3 WELD RESULT / SAVING WELD ALARMS

The welding results and alarms for the latest 100 spots are output to a text file. The file name is iwdg.dg. You can use the following three methods to save the data. The data output with the three methods is the same.

- Save data using the File screen.

Operation

MENUS

 → File → F4 [Backup] → [Select All of above] or [Diagnostic].

- Save data using the IWC alarm history screen (See section 4.3)
- Get data to a PC via FTP.

The contents of iwdg.dg are shown below.

Following the header line, parameters are listed in CSV format.

[illegible]

For easy to see, you paste this on to a spreadsheet as following.																		
PART ID	WELD ID	SCHEDULE No	WELD CURRENT (A)	Omitted	EXPANSION	FIXED ARM STRAIN	MOVABLE ARM STRAIN	TIME	CURRENT LOW	CURRENT HIGH	HEAT % LOW	HEAT % HIGH	LINE VOLT LOW	Omitted	W-CUR PRO GRA M LOW	W-CUR PRO GRA M HIGH	TOTAL W-TIME LONG	
0	1	1	2000			0	0	0	06-DEC-01 09:41	0	0	0	0		0		0	0
0	2	2	0		0	0	0	06-DEC-01 09:41	0	0	0	0	0		1	0	0	
0	1	3	0		0	0	0	06-DEC-01 09:42	0	0	0	0	0		0	0	0	
0	2	4	2000		0	0	0	06-DEC-01 09:42	0	0	0	0	0		0	0	0	
JOB123		1	5	2000		0	0	0	06-DEC-01 09:42	1	0	0	0	0		0	0	0
JOB123	pos2		6	2000		0	0	0	06-DEC-01 09:42	0	1	0	0	0		0	0	0
JOB123		1	7	2000		0	0	0	06-DEC-01 09:42	0	0	0	0	1		0	0	0
JOB123		2	8	2000		0	0	0	06-DEC-01 09:42	0	0	0	0	0		0	0	0
JOB123		1	9	2000		0	0	0	06-DEC-01 09:42	0	0	1	0	0		0	0	0
JOB123		2	10	2000		0	0	0	06-DEC-01 09:43	0	0	0	1	0		0	0	0
STYLE2		1	11	2000		0	0	0	06-DEC-01 09:43	0	0	0	0	0		0	0	0
STYLE2	Corner1		12	2000		0	0	0	06-DEC-01 09:43	0	0	0	0	0		0	0	0
STYLE2		1	13	2000		0	0	0	06-DEC-01 09:43	0	0	0	0	0		0	0	0
STYLE2		2	14	2000		0	0	0	06-DEC-01 09:43	0	0	0	0	0		0	0	0



Welding data



Alarms

The data cannot be saved in the controlled start menu or in the status in which no timer board is installed.

NOTE

File messages

- 1 "This file is not available at controlled start."
An attempt was made to save data in the controlled start mode. The data is not saved.
- 2 "No IWC(DG) board"
An attempt was made to save data in the status in which no timer board was installed. The data is not saved.

7.4 MONITOR OF WELDED

Welded is monitored as described below.

Monitoring for item (3) must be done using the PLC of the user.

(1) Monitoring by the timer: SPOT-063 NO CURRENT

If the weld current is not more than 5% of the setting, or thyristor ignition control fails, a stop occurs with the above alarm. Refer to alarm code 500 (NO CURRENT) in the manual, issued by DENGSHA MFG. CO., LTD. for more details.

(2) Monitoring by the robot: SPOT-063 No Weld Detected

If the timer welding results indicate that the Line volt is 0 although the Weld/NO WELD signal is on, a stop occurs with the above alarm.

(3) Monitoring by the PLC (No Weld detection signal)

The no weld detection signal can be used by the following settings:

\$IWCDGDBG[62]: Pulse width [msec]

(Default 500: signal level held constant by setting 0)

\$IWCDGDBG[64]: DO[] signal number

(Default 0: this function is unused)

After the setting is changed, the power must be turned off then back on.

When the Weld/NO WELD cell input signal has been set to off by simulation (the operator has forgotten to change UNSIM the signal), the robot and timer can not check the actual weld mode in PLC. In this case, PLC can check the weld result from timer with using this no weld detection signal.

If the Line volt in the welding results feed back from the timer board is 0, the robot outputs a pulse signal specified as mentioned \$IWCDGDBG[64].

When the PLC's weld mode is enable status and the above signal is set to on, the PLC inputs the hold signal to the robot. The hold signal stops the robot.

When the robot is stopped by the hold signal from the PLC, the program execution has gone ahead of the spot instruction. Before restart, it is necessary to change the program start line to the spot instruction line. If an error in 1 or 2 above occurs, it is not necessary to change the start line.

The user have to design this checking PLC program and set the above signal. Before using this function, set the signals used to off.

7.5 WELD/NO WELD

When teach pendant is enabled, weld mode can be enabled and disabled on the soft panel screen.

Operation

MENUS → TEST CYCLE → F1 [TYPE] → Spot Weld

SPOT WELD					
					1/9
1	Gun operation:			STROKE	
2	Weld controller mode:			WELD	
3	Tryout mode:			DISABLED	
4	Weld IO sim:			DISABLED	
5	Weld Duration (ms):			500 ms	
6	Inform output to plc			DO[0]	
7	Force process complete:			DISABLED	
Servo Gun Motion in Single Step					
8	Pressure Motion:			ENABLED	
9	Stop @ Taught Pos:			ENABLED	
	[TYPE]			WELD	NOWELD >

Move the cursor on the welding enable/disable indication.

NOTE

When the Gun operation is disabled, the Weld controller mode cannot be WELD. When the ON/OFF key on the teach pendant is set to the OFF position, you cannot operate any items on the test cycle. In production, the "Weld/NO WELD" input signal of the cell interface takes priority.

Refer to the R-30iA Controller Spot Tool+ operator's manual (B-83124EN-1) or R-30iB Controller Spot Welding Function operator's manual (B-83284EN-4) for more details.

When teach pendant is disabled (at production time), welding can be enabled or disabled by the "welding enable/disable" input signal of the cell interface. If this signal is set to on, welding is enabled, if the signal is set to off, welding is disabled.

Operation

MENUS → I/O → F1 [TYPE] → Cell Interface

The screen display is changed to the input signal screen by F3, IN/OUT.

I/O Cell Inputs					
					1/5
	INPUT SIGNAL	TYPE #	SIM	STATUS	
1	Weld/NO WELD	DI[0]	U	ON	
2	Stroke/NO STROKE	DI[0]	U	***	
3	Rmt wtr svr rese	DI[0]	U	***	
4	Ret hom frm poun	DI[0]	U	***	
5	Tryout Mode	DI[0]	U	***	
	[TYPE]		IN/OUT	ON	OFF >

Simulated welding

To execute a spot instruction in simulation mode when no internal timer board is installed, use the Weld IO sim on the soft panel screen.

Operation

MENUS → TEST CYCLE → F1 [TYPE] → Spot Weld

SPOT WELD		1/9
1 Gun operation:	STROKE	
2 Weld controller mode:	NOWELD	
3 Tryout mode:	DISABLED	
4 Weld IO sim:	ENABLED	
5 Weld Duration (ms):	500 ms	
6 Inform output to plc	DO[0]	
7 Force process complete:	DISABLED	
Servo Gun Motion in Single Step		
8 Pressure Motion:	ENABLED	
9 Stop @ Taught Pos:	ENABLED	
[TYPE]		ENABLED DISABLED >

Set the Weld controller mode to the NOWELD.

Set the Weld IO sim to the ENABLED. By this setup, the welding machine is not started for welding.

Set the Weld duration. This item specifies the duration of the weld you want to simulate. The robot will close the gun, and wait for a period of time that is equal to the weld duration. After the robot has simulated the weld, it will move to the next point in the welding sequence, and simulate another weld, until the weld sequence has been completed.

NOTE

When the Weld controller mode is WELD, it is impossible to enable Weld IO sim. Refer to the R-30iA Controller Spot Tool+ operator's manual (B-83124EN-1) or R-30iB Controller Spot Welding Function operator's manual (B-83284EN-4) for more details.

8

RESTRICTIONS

- This software option can control only integrated timers manufactured by DENGENSHA MFG. CO., LTD.
- During execution of welding, any weld parameters cannot be written onto the welding timer board. For example, no value can be changed on the DEP(DG) Setup/schedule screen during welding execution.
- Do not change any system variable related to the integrated timer using a parameter instruction.
- You cannot use multiple programs simultaneously to perform spot welding.
- Integrated weld controller and digital weld controller can not be used together in a robot controller.
- The weld schedules (schedule number) to a integrated timer board cannot inform to digital/group signals.
- This software option does not support any dual guns. Gun change is supported.

9 OTHER NOTES

- Be sure to set a value longer than the welding time for “Weld complete time-out” on the spot config screen. After changing the value of the welding parameter, perform actual welding to confirm that the Weld complete time-out setting is sufficiently longer than the actual welding time.
- The weld schedules and settings can be updated via FTP (iwdgsche.sv and iwdgconf.sv can be put via FTP) only in the controlled start mode.
- In gun change system, the weld schedule date is needed by each guns because stepper data and trans turns ratio are set for each schedule.
(Example)
Schedule 1-50 for Gun1
Schedule 51-100 for Gun2
- If there is no need to distinguish stepper and counter and all guns have the same turns ratio, those guns can share the common weld schedules.
- After changing language, need to power cycle.

10 ALARMS

This chapter explains alarms related to the integrated timer.

SPOT-008 PAUSE Weld complete timeout

[Cause]

A welding completion time-out occurred. An error may occur during welding operation.

If no alarm was issued from the welder, the welding time may be longer than the time-out.

[Remedy]

On the alarm history screen, check whether a welding alarm was issued. If any, remove the cause.

If the time out value (def: 2000ms) is shorten, you can adjust with “Weld complete time-out” on the spot config screen.

SPOT-010 PAUSE Major alarm detected

[Cause]

A fault-level alarm was issued.

[Remedy]

On the alarm history screen (MENUS → Alarm → F3 HIST), check the alarms issued before this alarm.

SPOT-011 WARN Minor alarm detected

[Cause]

A warning-level alarm was issued.

[Remedy]

On the alarm history screen (MENUS → Alarm → F3 HIST), check the alarms issued before this alarm.

SPOT-046 WARN Weld auto retried %s time(s)

[Cause]

An alarm was issued from the welding machine and the indicated number of retries was performed.

[Remedy]

This message notifies you that retries were performed.

When retry is successful, the program does not stop.

Check the alarms issued before retries.

SPOT-061 WARN Request Failed

[Cause]

A request to the timer board failed.

[Remedy]

This alarm may be issued after a parameter is written.

It may be issued during welding. Retry while no welding is performed.

SPOT-063 STOP.G “No Weld Detected”

[Cause]

The Line Volt in monitor data was 0 in the welding result from the timer board even though the WELD/No Weld signal from the PLC is on.

[Remedy]

Please contact the weld timer maker for reason of Line Volt is 0.

SPOT-063 STOP.G “alarm-message”

[Cause]

FAULT alarm message is received from weld timer board.

[Remedy]

Refer to weld timer manual for this message.

SPOT-064 WARN “Weld request not executed.”

[Cause]

A welding request was rejected. The timer board was not ready for start welding.

[Remedy]

Refer to alarms on DEP(DG)/Alarm. Issue reset on this screen.

SPOT-064 WARN “alarm-message”

[Cause]

ALARM message is received from weld timer board.

[Remedy]

Refer to weld timer manual for this message.

SPOT-066 WARN Iwc NOT ready

[Cause]

No timer board is installed.

The software on the integrated timer has not started up.

Communication with the timer board is stopped.

[Remedy]

Check that a timer board is installed.

This alarm may occur for a few seconds after cold start. Wait a while, and you can reset this alarm.

When causes above 2 are removed and this alarm occurs during any operations or instructions, the cause is above 3.

Check the alarm history, and turn the power off, then on again.

SPOT-067 PAUSE No Shunt Trip output for WC:weld controller-number

[Cause]

Shunt trip DO is not set.

[Remedy]

Check the number set for Shunt trip DO on the Weld Interface I/O screen.

If your contactor does not have shunttrip interface, please change following system variables and power cycle.

\$IWPCFG.\$SHUNT_ENABLE[1]=FALSE WC:1

\$IWPCFG.\$SHUNT_ENABLE[2]=FALSE WC:2

SPOT-350 WARN IWC board not found

[Cause]

No IWC board is installed.

[Remedy]

Install an IWC board

SPOT-381 WARN IWC(DG) system error(%d)

[Cause]

A problem occurs in communication with the timer board.

[Remedy]

Write down the number and contact a service representative. Possible causes are listed below.

Number(s)	Possible cause
1,2, 13	The temporary pool area may be almost exhausted.
3	The timer board could not be initialized successfully.
4	A parameter could not be written to the timer board successfully.
5,6,7,8,9	A parameter could not be read from the timer board.
12	A counter value could not be written to the timer board successfully.

SPOT-382 PAUSE IWC(DG) illegal stepper number(%s)

[Cause]

An invalid gun series number (SN) is specified in RESET STEPPER[WC,SN,SV].

[Remedy]

Set 1 to 4.

SPOT-383 PAUSE IWC(DG) illegal contactor status(%d)

[Cause]

Parameter in CONTACTOR instruction is invalid.

[Remedy]

Enter a correct parameter into the CONTACTOR instruction.

SPOT-384 PAUSE IWC(DG) leak input (#)

[Cause]

The Leak Input signal is on.

[Remedy]

1. Remove the cause of the leakage.
2. Check that OFF is displayed for Leak input on the Weld Interface I/O screen.
3. Perform a reset.

If the Leak Input signal is still on after action 2 is taken, check wiring.

Check whether the polarity of the input signal is reversed.

To disable this check, set the signal number for Leak input to 0 on the Weld Interface screen, and turn the power off, then on again.

SPOT-385 PAUSE IWC(DG) illegal counter specified(#)

[Cause]

An invalid counter value (SV) is specified in RESET STEPPER[WC,SN,SV].

[Remedy]

Set a value between 0 and 2.

SPOT-386 PAUSE Reset stepper timeout %s

[Cause]

RESET STEPPER was executed, but a time-out period had elapsed before response was received from the welding timer board.

[Remedy]

If a SPOT-387 to 390 alarm may is issued, there is a problem in communication with the timer board. Turn the power off, then on again.

The time-out period for RESET STEPPER may be too short.

If a small value is specified for "Rst weld cntler time-out" in SETUP/Spot config, specify a larger value.

SPOT-387 WARN IWC(DG) timer no response

[Cause]

No response was received from the welding timer board. The requested operation could not be executed.

[Remedy]

Turn the power off, then on again.

SPOT-388 WARN IWC(DG) timer acc stat err(%d)

[Cause]

Because the access status of the welding timer board remains on, the requested operation cannot be performed.

[Remedy]

Turn the power off, then on again.

SPOT-389 WARN IWC(DG) timer exe flag err(#)

[Cause]

Because a welding timer board execution flag remains on, the requested operation cannot be performed.

#

- 1 The SYSRDY signal is on.
- 2 The data read flag is on.
- 3 The data write flag is on.
- 4 The function name read flag is on.
- 5 The counter set/reset flag is on.
- 6 The welding flag is on.
- 7 The data command execution flag is on.

[Remedy]

Turn the power off, then on again.

SPOT-390 WARN IWC(DG) timer bad response

[Cause]

An error occurred in the interface with the welding timer board.

[Remedy]

Contact a service representative. Check whether a software or hardware setting of the welding timer board was changed before the error occurred.

SPOT-391 PAUSE IWC(DG) timer memory error

[Cause]

An error occurred in the hardware system saved in memory in the welding timer board.

[Remedy]

Turn the power off, then on again. The hardware data in timer board is initialized, so readjustment is required. (See Subsection 2.3.5, "HARD (Hardware).")

For adjustment data, contact DENGENSHA MFG. CO., LTD.

SPOT-392 SYSTEM IWC(DG) timer not running

[Cause]

An error occurred because the welding timer board had not started.

[Remedy]

*systdy from the timer board is not turned off.

Contact DENGENSHA MFG. CO., LTD.

SPOT-393 PAUSE IWC(DG) timer support from Gun1 to Gun4

[Cause]

Gun number in RESET STEPPER instruction is invalid.

[Remedy]

Enter a correct gun number into the RESET STEPPER instruction

SPOT-394 WARN IWC(DG) drv sys error(%d)

[Cause]

A system error occurred during processing of communication with the timer board.

[Remedy]

Check the number at the end of the error message.

Smaller than 500	Write down the number and contact a service representative.
500 or greater	Check the I/O settings such as Shunt trip I/O output and stepper ready I/O output. Check whether a value outside the range is specified as the I/O type or number.

SPOT-395 STOP IWC(DG) timer now in weld

[Cause]

An attempt was made to start welding during welding.

[Remedy]

The weld time-out period is not sufficiently long as compared with the welding time.

Set a long value for "Weld complete time-out" in SETUP/Spot config as compared with the welding time.

SPOT-396 STOP IWC(DG) timer com err %d(%x)

[Cause]

The system is unable to communicate with the timer board.

[Remedy]

Check alarms posted with this alarm and do their remedies.

And inform it to a service representative.

When the first number is 100 af following,

SPOT-396 STOP IWC(DG) timer com err 100(Board id)

Check your timer board [SEL] Dip switch T0 T1.

For timer board 1, T0=0 T1=0.

For timer board 2, T0=1 T1=0.

SPOT-481 SYSTEM IWC(DG) Can't read system_ready(%d)

[Cause]

A system error occurred.

[Remedy]

Turn off and back on the power. If the problem persists, check the mounting state of the internal timer board.

SPOT-482 SYSTEM IWC(DG) Failed to set Weld Current flag(%d)

[Cause]

A system error occurred.

[Remedy]

Turn the power off and back on. If the problem persists, check the mounting state of the internal timer board. Pull out the timer board and insert it again.

SPOT-483 PAUSE IWC(DG) Board number duplicated(%d)

[Cause]

The board number of the internal timer board is overlapped.

[Remedy]

Check your timer board [SEL] Dip switch T0 T1.

For timer board 1, T0=0 T1=0.

For timer board 2, T0=1 T1=0.

SPOT-484 WARN IWC(DG) MB Gun no out of range(%d)

[Cause]

When the monitor box is used, the series number from the timer board is incorrect.

[Remedy]

Contact a service representative with the value in parentheses, the version number of the robot software, and the version number of the timer software.

SPOT-485 SYSTEM IWC(DG) DPRAM error (%d)

[Cause]

A system error occurred. An error occurred while writing to DPRAM. The timer board may be incorrectly installed.

[Remedy]

Pull out the timer board and insert it again. If the problem persists, a board number is indicated in parentheses after SPOT-485. At the same time, detailed information is indicated after SPOT-486. Write down these values and contact a service representative.

SPOT-486 SYSTEM %x,%x,%x

[Cause]

This is a status message for SPOT-485.

[Remedy]

N/A

SPOT-487 PAUSE IWC(DG) shunt trip DO error(%d)

[Cause]

The shunt trip signal is improperly set. A board number is indicated in parentheses.

[Remedy]

Check if the Shunt trip signal number is correct on the SPOT Weld interface I/O output screen. Turn the power off, then on again after changing.

SPOT-488 PAUSE IWC(DG) stepper cmp DO error(%d)

[Cause]

The stepper completion output DO is improperly set.

[Remedy]

The improper setting number is indicated in parentheses. Check the following system variable.

\$IWCPCFG.\$STPC_OT_IDX[]

Turn the power off, then on again after changing.

SPOT-489 PAUSE IWC(DG) DPRAM out DO error(%d,%d)

[Cause]

The DO of the DPRAM output function is improperly set.

[Remedy]

The first number in parentheses is the board number of the board on which an error occurred. The second number is the improper setting number. Change the setting with reference to the I/O alarm indicated together. Turn the power off, then on again after changing.

SPOT-490 WARN IWC(DG) timer now in weld %d

[Cause]

A welding timer board reset was attempted during welding.

[Remedy]

It is not allowed to make a welding timer reset during welding. Make a reset after completion of welding.

SPOT-491 PAUSE IWC(DG) No Weld detection DO error

[Cause]

DO index (\$IWCDGDBG[64]) for No Weld detection signal is except 0 and No Weld detection signal was fails to output.

[Remedy]

Check whether DO index (\$IWCDGDBG[64]) is assigned correctly.

Turn the power off, then on again after changing.

SPOT-492 STOP IWC(DG) PrmWriteErrWC%d S%d %d

[Cause]

The system was not able to write the parameter to the timer board. The first number indicates the weld controller number. The second number indicates the sequence number as follows.

S0: Common, S1-S255: Schedule, S256: Special function, S257-S260: GUN1-4.

The third number indicates the error code number.

[Remedy]

When error code is 2, the value of the parameter is outside the range. Inquire the value of the parameter of the weld controller maker. Set the system variable to a correct value according to the parameter.

WC1 S0:\$IWCDGCON[1].\$PARM[Parameter number]

WC1 S1-S255:\$IWCDGS[Schedule number].\$PARM[Parameter number]

WC1 S256:\$IWCDGCON[2].\$PARM[Parameter number]

WC1 S257-S260:\$IWCDGCON[3-6].\$PARM[Parameter number]

WC2 S0:\$IWCDGCON[7].\$PARM[Parameter number]

WC2 S1-S255:\$IWCDGS2[Schedule number].\$PARM[Parameter number]

WC2 S256:\$IWCDGCON[8].\$PARM[Parameter number]

WC2 S257-S260:\$IWCDGCON[9-12].\$PARM[Parameter number]

Call your FANUC or FANUC Robotics technical service representative if error code is not 2.

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REVISION RECORD

Edition	Date	Contents
03	Oct.,2012	Output signal to external equipment is added to subsection 2.2.3.
02	Jun,2012	Supported R-30iB controller's manual
01	Mar., 2007	

B-82624EN/03



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